University of Minnesota Micro Notes

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Contents

1 Introduction

This document serves as a collection of concepts covered in graduate level microeconomics. Specifically, the PhD Microeconomics sequence ECON 8101 (Werner), ECON 8102 (Phelan), ECON 8103 (Rustichini), and ECON 8104 (Rahman) at the University of Minnesota. If you ever encouter these notes and have questions or spot an error, be sure to reach out to me here. Note that everything below is very personalized. I will officially state some theorems but the main goal of writing everything out is to cement my own intuition. Most everything will include statements to help intimidating things feel a bit more friendly. Additionally, I speak a lot in the second person because these notes are written in a manner as to be read by myself in the past.

2 ECON 8101 (Werner)

The main units/concepts that have been covered thus far have been Production Sets, Production Functions, Profit Maximization, Convex Analysis and Duality, Supply and Profit, Preferences and Utility Functions, Walrasian Demand, the Slutsky Matrix, Revealed Preference, Theorem of Topkis. I will discuss the main ideas and theorems that pervade the aforementioned concepts.

2.1 Production

Definition 1 (Supermodularity)**.** *A function f is supermodular if for any x, y we have that*

$$f(x \vee y) - f(x) \geq f(x \wedge y) - f(y).$$

This also implies the complementary of inputs (which intuitively is something that is indeed defined by the production function and not an inherent chracteristic of the inputs themselves).

There are two more key concepts in this section: Profit Maximization and Cost Minimization. The concepts are simple enough and likely won't come up much on the midterm so I'm just going to note down the two relevant definitions.

Definition 2 (Profit Function)**.** *Given a price vector $p \in \mathbb{R}^n$, the profit function π^* is*

$$\pi^*(p) = \sup_{y \in Y} py.$$

Note here that Y is a production set which means it is usually structured as a vector of inputs and one ouput stacked on top of eachother where the inputs are usually normalized to be negative.

Definition 3 (Cost Function)**.** *Given a vector of prices $p \in \mathbb{R}^n_+$, output level $z \in \mathbb{R}$, and production function f, the cost function C^* is*

$$\begin{aligned} C^*(w, z) = \min_{x \geq 0} &\quad px \\ s.t. &\quad f(x) \geq z \end{aligned}$$

The conditional factor demand correspondence x^* is defined as $C^*(p, z) \equiv px^*(p, z)$ or can also be seen as

$$\begin{aligned} x^*(w, z) = \arg\min_{x \geq 0} &\quad px \\ s.t. &\quad f(x) \geq z \end{aligned}$$

I need to be comfortable with working with these optimization objects. They are not things that you can do simple algebra with but require more nuanced arguments that take advantage of inequalities and properties of the maxima and minima. There are often many problems regading these concepts, however, that try to make one reason about

2.2 Consumer Theory

Definition 4 (Indirect Utility Function). *The function u^* such that given a price vector $p \in \mathbb{R}^n_+$ and income $w \in \mathbb{R}_{++}$*

$$u^*(p, w) = \max_{x \geq 0} \quad u(x)$$
$$s.t. \quad px \leq w$$

Definition 5 (Walrasian Demand Function). *The function x^* such that given a price vector $p \in \mathbb{R}^n_+$ and income $w \in \mathbb{R}_{++}$*

$$x^*(p, w) = \arg\max_{x \geq 0} \quad u(x)$$
$$s.t. \quad px \leq w$$

Note that $u^*(p, w) = u(x^*(p, w))$.

I want to dedicate a small section to talk about the Walrasian demand function and the indirect utility function. The key intuition to keep in mind is Walras law, that given prices and income, the optimal consumption bundle consumes all of the income which is something that is guaranteed when a utility function is locally non-satiated. This is a necessary condition of a bundle that optimized utility, not sufficient. It is only a relation between optimal bundles and their corresponding prices not that a bundle that satisfies that price relation is optimal. This is something that often trips me up. If we want to make statements about similar consumption bundles by price then we have to use the definition, not the price relation that lns gives us.

Theorem 2.1 (Generalized Weak Axiom of Revealed Preference - GWARP). *Given two price vectors $p^a, p^b \in \mathbb{R}^n_{++}$ and consumption bundles $x^a, x^b \in \mathbb{R}^n_+$. If we have that*

$$p^a x^b \leq p^a x^a$$

then

$$p^b x^b \leq p^b x^a$$

follows. Tangent to this GWARP for Walrasian demand functions. Specifically, given prices $p, p' >> 0$ and incomes w, w' we have that

$$px^*(p', w') \leq w \implies p' x^*(p, w) \geq w'.$$

Intuitively we can think of this as analyzing the preference behavior of individuals by looking at what types of bundles they pick (hence the "revealed prefence"). For the Walrasian demand function, think of it as; if a given utility maximizing consumption bundle (under prices and income basket a) is affordable given prices and income basket b , then the utility maximizing consumption bundle under prices and income basket b is not affordable given prices and income basket a.

The strong axiom of revealed preference is an argument in transitivity. If we have a chain of consumption bundles whose preferences have been "revealed" then we can relate the first and $n-$th bundle in that chain.

Theorem 2.2 (The Law of Compensated Demand)**.** *Let x^* be a Walrasian demand function of a consumer with continuous and locally non-satiated utility function. Then for every $p >> 0, w > 0, p' >> 0, w' > 0$ such that $w' = p'x^*(p, w)$ we have that*

$$[x^*(p', w') - x^*(p, w)][p' - p] \leq 0.$$

To be completely honest, I'm lacking in my intuition here. I think something like: if the optimal consumption bundle is affordable at a different price set then either the other bundles prices are lower or the other bundle is smaller.

Definition 6 (The Slutsky Matrix)**.** *Given a differentiable Walrasian demand function x^* we define the Slutsky matrix by element as*

$$s_{k\ell} = \frac{\partial x_k^*(p, w)}{\partial p_\ell} + \frac{\partial x_k^*(p, w)}{\partial w} x_\ell^*(p, w).$$

Law of compensated demand implies that this matrix is negative semi-definite and the matrix is also symmetric (we are often asked to confirm these facts).

2.3 Comparative Statics

The main concept to keep in mind here is the Theorem of Topkis. The motivation here is that we can use this Theorem to make statements about the behavior of certain functions with respect to certain variables. For example, say we have some input demand function $x^*(p, w, t)$ that is a function of input prices, output prices, and some tax rate and we want to analyze the behavior (non-decreasing, non-increasing) with respect to a to any of the tax rate or inputs, we can apply Topkis. On onset, it personally feels a little specialized, but it has the flavor of something that is subtly very powerful.

Theorem 2.3 (Theorem of Topkis)**.** *Consider a function $f : X \times T \to R$ and the corresponding function*

$$\phi(t) = \arg\max_{x \in X} f(x, t).$$

We can say that ϕ is nondecreasing (nonincreasing) in t on S if

1. *S is a lattice (the $\vee$ and $\wedge$ operations are included in S).*

2. *$f(x, t)$ is supermodular in x for every t*

3. $f(x,t)$ *has nondecreasing (nonincreasing) difference in* t. *Aka for* $t' > t$ *and* $x' > x$ *we have*

$$f(x',t') - f(x',t) \geq (\leq) f(x,t') - f(x,t).$$

2.4 Uncertainty and Utility

Now we start discussing the interplay between preferences / utility functions and risk. Before that, there are a handful of key definitions to consider.

Definition 7 (State Separable Utility Representation). *We say that a preference* $\succeq$ *has a state-separable utility representation, if there exist utility functions* $v_s : \mathbb{R}_+ \to \mathbb{R}$ *for all* s, *such that*

$$c \succeq \iff \sum_{s=1}^{S} v_s(c_s) \geq \sum_{s=1}^{S} v_s(c'_s),$$

for every $c, c' \in \mathbb{R}^S_+$.

Definition 8 (Expected Utility Representation). *We say that* $\succeq$ *has an expected utility representation with respect to probabilities* $\{\pi_s\}$, *if there exists a function* $v : \mathbb{R}_+ \to \mathbb{R}$ *such that*

$$c \succeq c' \iff \sum_{s=1}^{S} \pi_s v(c_s) \geq \sum_{s=1}^{S} \pi_s v(c'_s),$$

for every $c, c' \in \mathbb{R}^S_+$. *We call the function* v *in this case as the* <u>*von Neumann-Morgenstern (or Bersnoulli) utility*</u>

I'm not going to cover proofs here that won't be incredibly focused on during the prelims. One important theorem to consider, however, is the existence of a state separable utility representation.

Definition 9 (Axiom of Irrelevance of Common Consequences). *Take any* $c \in \mathbb{R}^S_+$ *and* $y \in \mathbb{R}$ *and write*

$$c_{-s}y = (c_1, c_2, \ldots, y, c_{s+1}, c_{s+2}, \ldots, c_S).$$

That it is, $c_{-s}y$ *is the vector* c *that has replaced* c_s *with* y. *A preference relation satisfies the ICC axiom if for all* $c, d \in \mathbb{R}^S_+$ *and any* $y, w \in \mathbb{R}_+$ *the following condition holds*

$$c_{-s}y \succeq d_{-s}y \iff c_{-s}w \succeq d_{-s}w.$$

Theorem 2.4 (Theorem 16.1). *If* $S \geq 3$ *and the preference relation* $\succeq$ *is strictly increasing and continuous. Then* $\succeq$ *has a state-separable utility representation iff it satisfies the ICC axiom.*

Now consider probabilities of states $\{\pi_s\}$such that $\pi_s > 0$ for each s. Additionally, we denote $E(c) = \sum_s \pi_s c_s$ as the expected value of $c = (c_1, \ldots c_S)$ and $\mathbf{E}(c) = (E(c), \ldots, E(c))$ as corresponding risk-free consumption plan.

Definition 10 (Risk Aversion). *We say that preference relation* $\succeq$ *is risk averse with respect to* $\{\pi_s\}$, *if*

$$\mathbf{E}(c) \succeq c$$

for every c.

Theorem 2.5 (Theorem 16.2)**.** *If $S \geq 3$ and the preference relation $\succeq$ is strictly increasing and continuous. Then $\succeq$ satisfies the ICC Axiom and is risk averse with respect to probabilities $\{\pi_s\}$ if and only if it has a concave expected utility representation with respect to $\{\pi_s\}$.*

Definition 11 (Ellsberg Paradox)**.** *An urn has 90 balls of which 30 are red and the rest are blue and yellow. Exact numbers of blue balls and yellow balls are not known. Consider bets of \$1 on a ball of a certain color (or colors) drawn from the urn. Denote bets by 1_R, 1_B, $1_{R\vee Y}$, etc. Typical preferences over bets are*

$$1_R \succ 1_B, \qquad 1_{B\vee Y} \succ 1_{R\vee Y}$$

This pattern of preferences is incompatible with expected utility: it cannot be that $\pi(R) > \pi(B)$ and $\pi(B \vee Y) > \pi(R \vee Y)$, because $\pi(B \vee Y) = \pi(B) + \pi(Y)$ holds for any probability measure π.

The intuition here is that there are various situations where common characteristics like transitivity and separability for preferences over lotteries fail because humans have preferences that tend to be risk averse. As such, in an attempt to write more consistent utility representations, we employ the use of something like below.

Definition 12 (Multiple-Prior Expected Utility)**.** *An alternative to expected utility and one that can explain the Ellsberg paradox is the **multiple-prior expected utility.** It takes the form*

$$\min_{P\in\mathcal{P}} E_P[v(c)], \tag{mpeu}$$

where $v : R_+ \to R$ is the von Neumann–Morgenstern utility (with no date-0 consumption) and $\mathcal{P}$ is a convex and closed set of probability measures on $\mathcal{S}$.

*The set of probability measures $\mathcal{P}$ reflects the agent's **ambiguous beliefs**.*

Next, let's try to tie together probabilities and preferences directly. Let $Z = \{z_1, \ldots, z_K\}$ be a finite set of outcomes. A **lottery** is a probability distribution on Z. It is an assignment of probabilities $\{p_i\}_{i=1}^K$ where p_i is the probability of winning outcome z_i. Agents can have pereferences over lotteries and

Definition 13 (Expected Utility Representation)**.** *Preference relation $\succeq$ on the set of lotteries $\mathcal{L}$ has an expected utility representation if there exists a function $v : Z \to \mathbb{R}$ such that*

$$L \succeq L' \iff \sum_{i=1}^K p_i v(z_i) \geq \sum_{i=1}^K p_i' v(z_i).$$

Definition 14 (Definition 18.1)**.** *$\tilde{z}$ first-order stochastically dominates $\tilde{y}$ if*

$$F_z(t) \leq F_y(t), \qquad \forall t \in [a, b]. \tag{1}$$

Theorem 2.6 (Theorem 18.2)**.** *$\tilde{z}$ first-order stochastically dominates $\tilde{y}$ if and only if*

$$E[v(\tilde{z})] \geq E[v(\tilde{y})]$$

for every nondecreasing continuous v.

That is, $\tilde{z}$ FSD $\tilde{y}$ if and only if every expected-utility maximizing agent with nondecreasing utility prefers $\tilde{z}$ to $\tilde{y}$.

Definition 15 (Definition 18.4)**.** *$\tilde{z}$ second-order stochastically dominates $\tilde{y}$ if*

$$\int_a^w F_z(t)\,dt \leq \int_a^w F_y(t)\,dt, \qquad \forall w \in [a,b]. \tag{2}$$

If $\tilde{z}$ SSD $\tilde{y}$, then $E(\tilde{z}) \geq E(\tilde{y})$. Also, if $\tilde{z}$ FSD $\tilde{y}$, then $\tilde{z}$ SSD $\tilde{y}$.

Theorem 2.7 (Theorem 18.5)**.** *$\tilde{z}$ second-order stochastically dominates $\tilde{y}$ if and only if*

$$E[v(\tilde{z})] \geq E[v(\tilde{y})]$$

for every nondecreasing concave continuous v.

That is, $\tilde{z}$ SSD $\tilde{y}$ if and only if every agent with risk averse nondecreasing expected utility prefers $\tilde{z}$ to $\tilde{y}$. If $\tilde{z}$ SSD $\tilde{y}$ and $\tilde{z}$ and $\tilde{y}$ have the same expectation $E(\tilde{z}) = E(\tilde{y})$, then we say that $\tilde{y}$ is **more risky** than $\tilde{z}$.

2.5 Recursive Utility

This section usually corresponds to one specific type of prelim problem (2.7.1): we are asked to define families of recursive utility functions, define dynamic consistency and then show a handful of properties of a family of recursive utility functions that can be represented with some aggregator funciton.

To start at the beginning, however, consider one of the most common representation of utility. The discounted utility function

$$\sum_{t=0}^{\infty} \beta^t u(c_t).$$

From here, we can define

Definition 16 (Continuation Utility Function)**.** *A function which assigns utility for every consumption plan starting at date-t, denoted by $c^t = (c_t, c_{t+1}, \ldots)$, for every $t \geq 1$. Specifically, it is*

$$U_t(c^t) = \sum_{\tau=t}^{\infty} \beta^{\tau-t} u(c_\tau).$$

There are several properties of continuation functions and collections of continuation functions that are worth discussing.

Definition 17 (Dynamic Consistency)**.** *A family of continuation functions is dynamically consistent if for every t, we have that for every $c, d \in \mathbb{R}_+^\infty$ such that $c_\tau = d_\tau$ for $\tau < t$*

$$U_0(c) \geq U_0(d) \Longleftrightarrow U_t(c^t) \geq U_t(d^t).$$

Definition 18 (Recursive Utility and Stationarity)**.** *More generally, a family of utility functions $\{U_t\}$ is recursive if*

$$U_t(c^t) = G(u(c_t), U_{t+1}(c^{t+1}))$$

for every c and t and some aggregator function $G : \mathbb{R}^2 \to \mathbb{R}$ and in-period utility function $u : \mathbb{R}_+ \to \mathbb{R}$

A recursive family of utility functions is stationary if $U_t = U$ for every t. That is

$$U(c^t) = G(u(c_t), U(c^{t+1})).$$

Definition 19 (Preference for Early Consumption)**.** *A recursive family of utility functions $\{U_t\}$ exhibits preference for early consumption if for every $c \in C$, $w, v \in \mathbb{R}_+$ such that $w > v$, and every t, it is the case that*

$$U_t((c_1, \ldots, c_{t-1}, w, v, c_{t+2}, \ldots)) > U_t((c_1, \ldots, c_{t-1}, v, w, c_{t+2}, \ldots)).$$

Werner also talks a little about recursive utility under uncertainty, but in my experience, this has never showed up on either a HW, Midterm, Final, or Prelim problem. The concept is generally, not all that different.

Now we must also define a finite set of possible sets at any given time S. The product set S^∞ represents the set of all possible sequences of these states over infinite time. Also consider some probability measure Π over S^∞.

Definition 20 (Discounted Expected Utility)**.** *As discussed above, it is a function which assigns utility for every consumption plan starting at date-t, denoted by $c^t = (c_t, c_{t+1}, \ldots)$, for every $t \geq 1$. Specifically, it is*

$$U_t(c^t) = \sum_{\tau=t}^{\infty} \beta^{\tau-t} \mathbb{E}_t[u(c_\tau)].$$

Definition 21 (State Dependent Recursive Utility and Stationarity)**.** *A state dependent utility function $U : S \times C \to \mathbb{R}$ is stationary recursive if it satisfies*

$$U(s_t, c^t(s^t)) = F(c_t(s^t), \mu_{s_t}(U(c^{t+1}))), \quad \forall s_t, t$$

for every $c \in C$, where $F : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}$ is the aggregator and $\mu_s : \mathbb{R}^S \to \mathbb{R}$ is the state dependent certianty equivalent.

Although it is quite unlikely that there will be a question related to Epstein-Zin-Weil preferences. This is quite relevant for a lot of work that is being done in macro.

Definition 22 (Epstein-Zin-Weil)**.** *A recursive utility satisfying*

$$U_t(c^t) = \left(c_t^{\alpha} + \beta[\mathbb{E}_t(U_{t+1}(c^{t+1}))^{1-\rho}]^{\alpha/(1-\rho)}\right)^{1/\alpha}$$

for $\alpha =\neq 0$ *and* $\rho \neq 1$.

2.6 Social Choice

This section is very tame for Werner relative to everything else. There are rarely any questions regarding it. Consider a world with

- X, a set of social alternatives
- N, a set of individuals ($N = \{1, 2, \ldots, n\}$).
- $\succeq_i \in \mathcal{R}$, a preference relation of individual i on X and $\mathcal{R}$ is the set of all such preference relations. Let $\mathcal{P}$ denote the subset of $\mathcal{R}$ such that there are no indifferences.

With these objects in hand, we can then define

Definition 23 (Social Welfare Function)**.** *A function* $F\mathcal{P}^n \to \mathcal{R}$ *that assigns a (social) preference relationto every profile of individual preferences. Specifically, we denote this as*

$$\succeq_F \equiv F(\succ_1, \ldots, \succ_n).$$

Arrow's theorem is one of the groundbreaking results in economics that everyone is familiar with. We will look at the proof more in depth in the Rahman section, but there is a quick statement of it. First consider the following two qualities that a social welfare function F may have

- (Pareto) For every $x, y \in X$ and every profile $(\succ_1, \ldots, \succ_n)$, if $x \succ_i y$ for all i, then $x \succ_F y$.
- (Independence of Irrelevant Alternatives, IIA) For every $x, y \in X$ and every pair of profiles $(\succ_1, \ldots, \succ_n)$ and $(\succ'_1, \ldots, \succ'_n)$, it holds

$$\text{if } x \succ_i y \text{ iff } x \succ'_i y \text{ for every } i\text{, then } x \succeq_F \text{ iff } x \succeq'_F y.$$

Theorem 2.8 (Arrow's Theorem)**.** *If there are at least three alternative (i.e.* $|X| \geq 3$*) and the social welfare function F is both Pareto and satisfies IIA, then F is dictatorial.*

That is, there exists some i^* *such that if* $x \succ_{i^*} y$ *then* $x \succ_F y$ *for every* $x, y \in X$ *and every profile* $(\succ_1, \ldots, \succ_n)$.

2.7 Exercises

I didn't cover optimal portfolio questions in the content above but refer to problems (2.7.2) for content on them.

2.7.1 Prelim: Spring 2025

Consider a family of utility functions $\{U_t\}_{t=0}^{\infty}$ defined on the set ℓ_+^∞ of bounded, positive consumption plans of a single good over infinite time horizon without uncertainty.

(a) Define what it means for the utility functions $\{U_t\}$ to be recursive. Define what it means for $\{U_t\}$ to be dynamically consistent.

Solution: $\{U_t\}$ is **recursive** if we can write

$$U_t(c_t) = G(u(c_t), U_{t+1}(c^{t+1})),$$

for every c and t and some aggregator function $G : \mathbb{R}^2 \to \mathbb{R}$ and period-utility function $u : \mathbb{R}_+ \to \mathbb{R}$.

It is **dynamically consistent** if for every t, and $c, d \in \mathbb{R}_+^\infty$ such that $c_\tau = d_\tau$ for any $\tau < t$,

$$U_0(c) \geq U_0(d) \Longleftrightarrow U_t(c^t) \geq U_t(d^t).$$

■

(b) Consider a stationary recursive utility function with aggregator

$$G(z_1, z_2) = z_1 + \beta \ln(z_2 + 1),$$

for $z_1 \geq 0$ and $z_2 \geq 0$, where $0 < \beta < 1$, and linear period-utility $u(x) = x$. Show that this recursive utility function is dynamically consistent.

Solution: Pick any arbitrary $t \geq 0$. Using this, select any arbitrary $c, d \in \mathbb{R}_+^\infty$ such that $c_\tau = d_\tau$ for any $\tau < t$. We can write for any $\tau < t$

$$\begin{aligned}
&U_\tau(c^\tau) \geq U_\tau(d^\tau) \\
\Longleftrightarrow\ & c_\tau + \beta \ln\left(U_{\tau+1}(c^{\tau+1}) + 1\right) \geq d_\tau + \beta \ln\left(U_{\tau+1}(d^{\tau+1}) + 1\right), \\
\Longleftrightarrow\ & \beta \ln\left(U_{\tau+1}(c^{\tau+1}) + 1\right) \geq \beta \ln\left(U_{\tau+1}(d^{\tau+1}) + 1\right), \\
\Longleftrightarrow\ & \ln\left(U_{\tau+1}(c^{\tau+1}) + 1\right) \geq \ln\left(U_{\tau+1}(d^{\tau+1}) + 1\right), \\
\Longleftrightarrow\ & U_{\tau+1}(c^{\tau+1}) + 1 \geq U_{\tau+1}(d^{\tau+1}) + 1, \\
\Longleftrightarrow\ & U_{\tau+1}(c^{\tau+1}) \geq U_{\tau+1}(d^{\tau+1}).
\end{aligned}$$

Since we used double implication statements the whole way, we can inductively use this fact to generally state that

$$U_s(c^s) \geq U_s(d^s) \iff U_k(c^k) \geq U_k(d^k),$$

for any $s, k \leq t$. Pick $s = 0$ and $k = t$ and this is exactly the definition of dynamic consistency for our utility function. ■

(c) A stationary recursive utility function U is said to exhibit preference for early consumption if, for any consumption plan that yields w at date t and y at date $t+1$, with $w > y$, the agent prefers it to an otherwise identical plan (i.e., consumption is the same at all other dates) that delivers y at date t and w at date $t+1$. State this definition formally. Show that the recursive utility function of part (b) exhibits preference for early consumption.

Solution: A recursive family of functions exhibits a preference for early consumption if for every $c \in C$, $w, y \in \mathbb{R}_+$ with $w > y$, and $t \geq 0$ we have that

$$U_t(\hat{c}^t) > U_t(\bar{c}^t)$$

where $\hat{c} = (c_1, \ldots, c_{t-1}, w, y, c_{t+2}, \ldots)$ and $\bar{c} = (c_1, \ldots, c_{t-1}, y, w, c_{t+2}, \ldots)$.

To prove this for the recursive utility function from before, pick any $c \in C$, $w, y \in \mathbb{R}_+$ with $w > y$, and $t \geq 0$. We can then write out

$$\begin{aligned}
&U_t(\hat{c}^t) > U_t(\bar{c}^t), \\
\iff &w + \beta \ln(U_{t+1}(\hat{c}^{t+1}) + 1) > y + \beta \ln(U_{t+1}(\bar{c}^{t+1}) + 1), \\
\iff &w + \beta \ln(\beta \ln(U_{t+2}(\hat{c}^{t+2}) + 1) + y + 1) > y + \beta \ln(\beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + w + 1).
\end{aligned}$$

From here we continue

$$\begin{aligned}
\iff &w - y > \beta \ln(\beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + w + 1) - \beta \ln(\beta \ln(U_{t+2}(\hat{c}^{t+2}) + 1) + y + 1), \\
\iff &1 > \frac{\beta \left[\ln(\beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + w + 1) - \ln(\beta \ln(U_{t+2}(\hat{c}^{t+2}) + 1) + y + 1)\right]}{w - y}, \\
\iff &1 > \frac{\beta \left[\ln(w + A) - \ln(y + A)\right]}{w - y},
\end{aligned}$$

where $A = \beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + 1 = \beta \ln(U_{t+2}(\hat{c}^{t+2}) + 1) + 1$. From here, using mean value theorem, we can write

$$\frac{\beta \left[\ln(w + A) - \ln(y + A)\right]}{w - y} < \frac{\beta}{w + A} \leq \beta < 1.$$

Since we used double implications the whole way, we have shown that

$$U_t(\hat{c}^t) > U_t(\bar{c}^t)$$

and indeed this family of utility functions exhibits preference for early consumption.

One note is that instead of using the mean value theorem, we can use the property that $\ln(1+x) < x$ for all $x > 0$. Specifically, we can write $B = \beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1)$, which we know is greater than 0, and then define $C_w = \beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + w$ and $C_y = \beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + y$. This let's us write both of the expressions

$$\begin{aligned} \ln(1 + C_w) < C_w &\implies \beta \ln(1 + C_w) < C_w \\ \ln(1 + C_y) < C_y &\implies \beta \ln(1 + C_y) < C_y \end{aligned}$$

Subtracting these two expressions gives

$$w - y > \beta[\ln(1 + C_w) - \ln(1 + C_y)],$$

which is exactly the expression we needed to prove above. ■

(d) Derive the marginal rate of intertemporal substitution between consumption at dates $t+1$ and t, and demonstrate that it is strictly less than 1 for every consumption plan.

Solution: Since this utility function is a recursive one and also linearly separable, we can pick any consumption plan $c \in C$ and write

$$\begin{aligned} MRS_{t+1,t} &= \frac{\partial U_t(c^t)/\partial c_{t+1}}{\partial U_t(c^t)/\partial c_t}, \\ &= \frac{\frac{1}{\partial c_{t+1}}\left[G(c_t, G(c_{t+1}, U_{t+2}(c^{t+2}))\right]}{\frac{1}{\partial c_t}\left[G(c_t, G(c_{t+1}, U_{t+2}(c^{t+2}))\right]}, \\ &= \frac{\frac{1}{\partial c_{t+1}}\left[c_t + \beta \ln(1 + G(c_{t+1}, U_{t+2}(c^{t+2})))\right]}{\frac{1}{\partial c_t}\left[c_t + \beta \ln(1 + G(c_{t+1}, U_{t+2}(c^{t+2})))\right]}, \\ &= \frac{1}{\partial c_{t+1}}\left[c_t + \beta \ln(1 + G(c_{t+1}, U_{t+2}(c^{t+2})))\right], \\ &= \frac{1}{\partial c_{t+1}}\left[\beta \ln(1 + G(c_{t+1}, U_{t+2}(c^{t+2})))\right], \\ &= \frac{1}{\partial c_{t+1}}\left[\beta \ln(1 + c_{t+1} + \beta \ln(U_{t+2}(c^{t+2}) + 1))\right], \\ &= \frac{\beta}{1 + c_{t+1} + \beta \ln(U_{t+2}(c^{t+2}) + 1)}, \end{aligned}$$

$$
\begin{aligned}
&\leq \frac{\beta}{1+c_{t+1}}, \\
&\leq \beta, \\
&< 1.
\end{aligned}
$$

■

2.7.2 Prelim: Fall 2024

There are two assets: a risk-free asset with return r_f and a risky asset with return $\tilde{r}$. Returns are per-dollar returns, meaning they represent the total return for one dollar invested. The agent has an expected utility function with von Neumann (or Bernoulli) utility index

$$v(x) = -(\alpha - x)^2,$$

for $\alpha > 0$, and initial wealth $w > 0$. Assume that $\alpha > wr_f$. Negative investment (i.e., short selling) is allowed for both assets.

(a) Find the optimal investment in the risky asset as a function of the expected return and the variance of the return on the risky asset, the risk-free return, and the agent's wealth.

Solution: Given wealth $w > 0$, the agent experiences the following maximization problem

$$\max_{w_r} \quad \mathbb{E}\left[v(r_f(w - w_r) + \tilde{r}w_r)\right],$$

which we can equivalently write as

$$\max_{w_r} \quad \mathbb{E}\left[v((r_f - \tilde{r})w_r + r_f w)\right].$$

Here is the corresponding first order condition

$$
\begin{aligned}
\mathbb{E}\left[(\tilde{r} - r_f)v'((\tilde{r} - r_f)w_r + r_f w)\right] &= 0, \\
\mathbb{E}\left[(-2)(\tilde{r} - r_f)(\alpha - (\tilde{r} - r_f)w_r - r_f w)\right] &= 0, \\
\mathbb{E}\left[(\tilde{r} - r_f)\alpha - (\tilde{r} - r_f)^2 w_r - r_f(\tilde{r} - r_f)w\right] &= 0, \\
\mathbb{E}[\tilde{r} - r_f]\alpha - \mathbb{E}[(\tilde{r} - r_f)^2]w_r - + r_f w\mathbb{E}[\tilde{r} - r_f] &= 0, \\
\frac{\mathbb{E}[\tilde{r} - r_f](\alpha - r_f w)}{\mathbb{E}[(\tilde{r} - r_f)^2]} &= w_r, \\
\frac{\mathbb{E}[\tilde{r} - r_f](\alpha - r_f w)}{\mathbb{E}[\tilde{r}^2 - 2\tilde{r}r_f + r_f^2]} &= w_r,
\end{aligned}
$$

$$\frac{\mathbb{E}[\tilde{r}-r_f](\alpha - r_f w)}{Var(\tilde{r}) + (\mathbb{E}[\tilde{r}]^2 - 2r_f\mathbb{E}[\tilde{r}] + r_f^2} = w_r.$$

Finally, we can write

$$\boxed{w_r = \frac{\mathbb{E}[\tilde{r}-r_f](\alpha - r_f w)}{Var(\tilde{r}) + \big(\mathbb{E}[\tilde{r}-r_f]\big)^2}.}$$

■

Consider the following comparative statics exercise. The return $\tilde{r}$ is changed to a more risky return $\tilde{r}'$. The expected return of $\tilde{r}'$ is the same as of $\tilde{r}$, that is, $E(\tilde{r}') = E(\tilde{r})$.

(b) Prove that $\mathrm{var}(\tilde{r}') \geq \mathrm{var}(\tilde{r})$, where var denotes the variance.

Solution: Formally, from the definition of a riskier asset, we know that

$$\mathbb{E}[v(\tilde{r}')] \geq \mathbb{E}[v(\tilde{r})].$$

Let's go from here then

$$\begin{aligned}
\mathbb{E}[v(\tilde{r}')] &\leq \mathbb{E}[v(\tilde{r})],\\
\mathbb{E}[-(\alpha-\tilde{r}')^2] &\leq \mathbb{E}[-(\alpha-\tilde{r})^2],\\
\mathbb{E}[(\alpha-\tilde{r}')^2] &\geq \mathbb{E}[(\alpha-\tilde{r})^2],\\
\mathbb{E}[\alpha^2 - 2\alpha\tilde{r}' + \tilde{r}'^2] &\geq \mathbb{E}[\alpha^2 - 2\alpha\tilde{r} + \tilde{r}^2],\\
\mathbb{E}[\tilde{r}'^2] &\geq \mathbb{E}[\tilde{r}^2],\\
\mathbb{E}[\tilde{r}'^2] - (\mathbb{E}[\tilde{r'}])^2 &\geq \mathbb{E}[\tilde{r}^2] - (\mathbb{E}[\tilde{r'}])^2,\\
\mathbb{E}[\tilde{r}'^2] - (\mathbb{E}[\tilde{r'}])^2 &\geq \mathbb{E}[\tilde{r}^2] - (\mathbb{E}[\tilde{r}])^2,\\
Var(\tilde{r}') &\geq Var(\tilde{r}).
\end{aligned}$$

■

(c) Suppose that $E(\tilde{r}) > r_f$. Show that the optimal investment in the risky asset with more risky return $\tilde{r}'$ is smaller than the optimal investment with less risky return $\tilde{r}$, all else unchanged.

Solution: We know from part (a) that the optimal investments in the less risky and more risky asset are respectively

$$w_r = \frac{\mathbb{E}[\tilde{r}-r_f](\alpha - r_f w)}{Var(\tilde{r}) + \big(\mathbb{E}[\tilde{r}-r_f]\big)^2}, \quad w_{r'} = \frac{\mathbb{E}[\tilde{r}'-r_f](\alpha - r_f w)}{Var(\tilde{r}') + \big(\mathbb{E}[\tilde{r}'-r_f]\big)^2}.$$

To make the comparison, consider the sequence of double implications

$$\begin{gathered}
w_r \geq w_{r'}, \\
\frac{\mathbb{E}[\tilde{r} - r_f](\alpha - r_f w)}{Var(\tilde{r}) + \left(\mathbb{E}[\tilde{r} - r_f]\right)^2} \geq \frac{\mathbb{E}[\tilde{r}' - r_f](\alpha - r_f w)}{Var(\tilde{r}') + \left(\mathbb{E}[\tilde{r}' - r_f]\right)^2}, \\
\frac{1}{Var(\tilde{r}) + \left(\mathbb{E}[\tilde{r} - r_f]\right)^2} \geq \frac{1}{Var(\tilde{r}') + \left(\mathbb{E}[\tilde{r}' - r_f]\right)^2}, \\
Var(\tilde{r}) + \left(\mathbb{E}[\tilde{r} - r_f]\right)^2 \leq Var(\tilde{r}') + \left(\mathbb{E}[\tilde{r}' - r_f]\right)^2, \\
Var(\tilde{r}) \leq Var(\tilde{r}').
\end{gathered}$$

Since the last line is true (as we showed in part (c)), it must be the case that $w_{r'} < w_{r'}$.

■

Suppose that the return $\tilde{r}$ second-order stochastically dominates $\tilde{r}'$ rather than being more risky. In this case, do not assume that the expected returns are equal.

(d) Does the argument from part (c), which shows that the optimal investment with return $\tilde{r}'$ is smaller than the optimal investment with $\tilde{r}$, still hold? Explain why or why not.

Solution: In this case the argument does not hold. It could be the case that the $\tilde{r}$ second-order stochastically dominates $\tilde{r}'$ but the expected return $\mathbb{E}[\tilde{r}']$ is so much higher than the expected return from the risk-less asset $\mathbb{E}[\tilde{r}]$ that it overcomes the agent's aversion to risk. ■

3 ECON 8102 (Phelan)

Phelan's class is very good for grounding intuition about competitive equilibrium. We don't cover as many topics as in the later mini's, which gives one a lot of time to really understand what's going on. Additionally, he doesn't publish any official set of notes, so there are likely going to be some holes.

3.1 Economy Primitives

Before we dive into the meat of the content of this course, we need to start with some basic primitives. I'll set up two types of economies as contextual environments and then dive into specifics about everything.

Pure Endowment Economies: Economy in which there exists only households who are bestowed an endowment of resources by some unseen power. There are

- A set of agents, $\mathcal{I} = \{1, \ldots, I\}$ for some $I \in \mathbb{N}$.
- A set of goods/commodities, $\mathcal{M} = \{1, \ldots, M\}$ for some $M \in \mathbb{M}$.
- Each agent i has a consumption set $C_i \subseteq \mathbb{R}^m$ which indicates the possible set of bundles that agent i can consume.
- Endowments of each good for each agent, $e = \{\{e_{i,m}\}_{m\in\mathcal{M}}\}_{i\in\mathcal{I}}$, $e_{i,m} \geq 0$.
- Allocation: consumption allocations for agents, $c = \{\{c_{i,m}\}_{m\in\mathcal{M}}\}_{i\in\mathcal{I}}$, $c_{i,m} \in \mathbb{R}$.

Production Economies: Economy similar to above but there are now also firms who are able to produce consumption goods as well.

- A set of agents, $\mathcal{I} = \{1, \ldots, I\}$ for some $I \in \mathbb{N}$.
- A set of firms, $\mathcal{H} = \{1, \ldots, H\}$ for some $H \in \mathbb{N}$.
- A set of goods/commodities, $\mathcal{M} = \{1, \ldots, M\}$ for some $M \in \mathbb{M}$.
- Each agent i has a consumption set $C_i \subseteq \mathbb{R}^m$ which indicates the possible set of bundles that agent i can consume.
- Each firm i has a production set $Y_h \subseteq \mathbb{R}^m$ which indicates the possible set of bundles that firm h can produce.
- Ownerships of person i in firm h: $\theta_{i,h} \in [0,1]$ for all i, h.
- Endowments of each good for each agent, $e = \{\{e_{i,m}\}_{m\in\mathcal{M}}\}_{i\in\mathcal{I}}$, $e_{i,m} \geq 0$.
- Allocation: tuple of consumption allocations and firm production plans,

  $$(c, y) = (\{\{c_{i,m}\}_{m\in\mathcal{M}}\}_{i\in\mathcal{I}}, \{\{y_{h,m}\}_{m\in\mathcal{M}}\}_{h\in\mathcal{H}})$$

  for $c_{i,m} \in \mathbb{R}$, $y_{h,m} \in \mathbb{R}$.

With the setups above in mind, here are a couple of definitions that will likely come up in problems.

Definition 24 (Utility Possibilities Set)**.** *The set of attainable vectors of utility levels. That is*

$$U = \left\{ \{u_i(c_i)\}_{i\in\mathcal{I}} :, c_i \in C_i, \sum_i c_{i,m} \leq \sum_i e_{i,m}, \forall m \right\}.$$

Definition 25 (Utility Possibilities Frontier)**.** *Let U be the utility possibilities set. The UPF is the set*

$$\left\{ \{u_i(c_i)\}_{i\in\mathcal{I}} \in U : \nexists (u_1', u_2') \in U \quad s.t. \quad u_i' > u_i, \forall i \right\}.$$

Definition 26 (Better than Sets)**.** *The Better than Set, $B(c)$, for a given consumption bundle c is the set of all consumption bundles that an agent prefers. [I NEED TO CHECK OVER THIS BECAUSE PHELAN ASKS FOR SOMETHING DIFFERENT IN THE HWS]*

Finally, we say that a specific allocation is **feasible** if the total amount of a specific good consumed across all agents is less than the sum of the endowment of that good across every agent (and production of that good across all firms). To put this in more specific terms, for endowment economies, an allocation c, given endowments e is feasible if $c_i \in C_i$ for all $i \in \mathcal{I}$ and for all $m \in \mathcal{M}$

$$\sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i} e_{i,m}.$$

For production economies, an allocation (c, y), given endowments e is feasible if if $c_i \in C_i$ for all $i \in \mathcal{I}$, $y_h \in Y_h$ for all $h \in \mathcal{H}$, and for all $m \in \mathcal{M}$

$$\sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i} e_{i,m} + \sum_{h \in \mathcal{H}} y_{h,m}.$$

3.2 Efficiency

You should not be surprised that we are once again talking about Pareto efficiency. It is, however, very straightforward in the Phelan barebones world.

Definition 27 (Pareto Dominating Allocation)**.** *An allocation (c, y) pareto dominates another allocation $(\hat{c}, \hat{y})$ if*

- $c_i \succeq_i \hat{c}_i$ *for all i*
- $c_j \succ_j \hat{c}_j$ *for at least one j*

Definition 28 (Pareto Efficient)**.** *An allocation is pareto efficient if it is feasible and there is no alternative feasible allocation that pareto dominates it.*

There two important problems that allow us to think about reaching efficient allocations from the perspective of a social planner. Both may or may not lead to the same outcomes. I will let the exercises below speak more to exporing the depth behind each of these problems and just define it here for now.

Definition 29 (Arrow / Planner's Problem (for pure endowment economies))**.** *The arrow / planner's problem for agent i, given $\bar{u}_j$ for all $j \in \mathcal{M}$ such that $j \neq i$, is the following maximization problem.*

$$\begin{aligned} \max_{c_i \in C_i} \quad & U_i(c_i) \\ s.t. \quad (1) \quad & \sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_{i,m}, \\ (2) \quad & U_j(c_j) \geq \bar{u}_j, \ \forall j \neq i. \end{aligned}$$

Definition 30 (Arrow / Planner's Problem (for production economies))**.** *The arrow / planner's problem for agent i, given $\bar{u}_j$ for all $j \in \mathcal{M}$ such that $j \neq i$, is the following maximization problem.*

$$\begin{aligned}
&\max_{c_i \in C_i} && U_i(c_i) \\
&s.t. && (1) \quad \sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_{i,m} + \sum_{h \in \mathcal{H}} y_{h,m}, \ \forall m \in \mathcal{M}, \\
& && (2) \quad U_j(c_j) \geq \bar{u}_j, \ \forall j \neq i.
\end{aligned}$$

Definition 31 (Negishi / Social Planner's Problem (for pure endowment economies))**.** *The negishi / social planner's problem, given weights λ_i for all $i \in \mathcal{I}$ is the following maximization problem*

$$\begin{aligned}
&\max_{c \in C} && \sum_{i \in \mathcal{I}} \lambda_i U_i(c_i) \\
&s.t. && \sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_{i,m}, \ \forall m \in \mathcal{M}.
\end{aligned}$$

Definition 32 (Negishi / Social Planner's Problem (for production economies))**.** *The negishi / social planner's problem, given weights λ_i for all $i \in \mathcal{I}$ is the following maximization problem*

$$\begin{aligned}
&\max_{c \in C} && \sum_{i \in \mathcal{I}} \lambda_i U_i(c_i) \\
&s.t. && \sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_{i,m} + \sum_{h \in \mathcal{H}} y_{h,m}, \ \forall m \in \mathcal{M}.
\end{aligned}$$

3.3 Equilibrium Concepts

My biggest takeaway from this class was thinking more deeply about what an equilibrium or solving an economy means. I personally come from a macro background and very much just accepted the fact that a competive equilibrium is nothing more than a set of objects that satisfy another set of conditions. It just so happens to be the generally agreed upon solution concept but there are many more types of solution concepts that we could define.

For example, consider a simple pure endowment economy. One equilibrium concept that we could define is an autarky equilibrium where every agent consumes their endowment and prices are set such that no trade occurs. This too can be considered an equilibrium, we just tend to focus on competitive equilibrium. One other equilibrium concept that Phelan talks a lot about though is what he calls Jungle Equilibrium. It is not likely to show up in a prelim problem, but it is still an interesting concept to think about.

Definition 33 (Jungle Equilibrium)**.** *Fix some ordering of agents from weakest to strongest. Specifically, an ordering S where $i \, S \, j$ means agent i is stronger than agent j where if $i \, S \, j$ then not $j \, S \, i$. An allocation such that no agent can assemble a more preferred bundle by combining his own bundle either with a bundle that is held by one of the agents weaker than him or with the bundle that is freely available. Formally, it is a consumption allocation $c = \{\{c_{i,m}\}_{i \in \mathcal{I}}\}_{m \in \mathcal{M}}$ such that*

- $c_i \in C_i$, *for all* $i \in \mathcal{I}$.
- $\sum_i c_{i,m} \leq e_m$, *for all* $m \in \mathcal{M}$.
- *For all* $i, j \in \mathcal{I}$ *such that* $i \, S \, j$. *We have that* $c_i \succeq_i \hat{c}_i$ *for all* $\hat{c}_i \in C_i$ *such that* $\hat{c}_{i,m} \leq c_{i,m} + c_{j,m}$ *for all* $m \in M$.

Beyond this, the focus of the class was on exploring competitive equilibria and some basic important concepts (such as interpreting goods in different time periods as inherently different goods, or seeing how ownership plays with preferences (3.5.1)) that each of the exercises below will address.

Definition 34 (Competitive Equilibrium (for pure endowment economies))**.** *Consider a pure endowment economy. A competitive equilibrium is a collection of allocations and prices* (c, p) *such that*

- $p \in \mathbb{R}^M_+$.
- *(Feasibility)* $c_i \in C_i$ *for all* i.
- *(Feasibility - Market Clearing)* $\sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_m$ *for all* $m \in \mathcal{M}$.
- *(Feasibility - Budget Constraint)* $\sum_{m \in \mathcal{M}} p_m c_{i,m} \leq \sum_{m \in \mathcal{M}} p_m e_{i,m}$ *for all* $i \in \mathcal{I}$.
- *(Optimality) For any* $i \in \mathcal{I}$, *if* $\hat{c}_i \succ c_i$ *then* $\sum_{m \in \mathcal{M}} p_m \hat{c}_{i,m} > \sum_{m \in \mathcal{M}} p_m e_{i,m}$

Definition 35 (Competitive Equilibrium (for production economies))**.** *Consider a production economy. A competitive equilibrium is a collection of consumption allocations, firm production plans, and prices* (c, y, p) *such that*

- $p \in \mathbb{R}^M_+$.
- *(Feasibility)* $c_i \in C_i$ *for all* i.
- *(Feasibility)* $y_h \in Y_h$ *for all* h.
- *(Feasibility - Market Clearing)* $\sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{h \in \mathcal{H}} y_h + \sum_{i \in \mathcal{I}} e_m$ *for all* $m \in \mathcal{M}$.
- *(Feasibility - Budget Constraint)* $\sum_{m \in \mathcal{M}} p_m c_{i,m} \leq \sum_{m \in \mathcal{M}} p_m e_{i,m} + \sum_{h \in \mathcal{H}} \theta_{i,h} \sum_{m \in \mathcal{M}} p_m y_{h,m}$ *for all* $i \in \mathcal{I}$.
- *(Optimality - Household) For any* $i \in \mathcal{I}$, *if* $\hat{c}_i \succ c_i$ *then* $\sum_{m \in \mathcal{M}} p_m \hat{c}_{i,m} > \sum_{m \in \mathcal{M}} p_m e_{i,m} + \sum_{h \in \mathcal{H}} \theta_{i,h} \sum_{m \in \mathcal{M}} p_m y_{h,m}$.
- *(Optimality - Firm) If* $\hat{y}_h$ *is such that* $\sum_{m \in \mathcal{M}} p_m \hat{y}_{h,m} < \sum_{m \in \mathcal{M}} p_m y_{h,m}$, *then* $\hat{y}_h \notin Y_h$.

As a final note, a decent chunk of Phelan's class was dedicated to proving the welfare theorems for the two different types of economies listed above. I'm just going to list them for now since he doesn't usually ask for proofs.

Theorem 3.1 (First Welfare Theorem (for pure endowment economies)). *Let preferences be locally non-satiatable for every agent. If (p,c) is a competitive equilibrium, then c is pareto efficient.*

Theorem 3.2 (Second Welfare Theorem (for pure endowment economies)). *Let preferences be continous, convex, and strongly monotone for every agent. If c is pareto efficient (where $c >> 0$), then there exists some e, p such that (p,c) is a competitive equilibrium.*

Theorem 3.3 (First Welfare Theorem (for production economies)). *Let preferences be locally non-satiatable for every agent. If (p,c,y) is a competitive equilibrium, then (c,y) is pareto efficient.*

Theorem 3.4 (Second Welfare Theorem (for production economies)). *Let preferences be continous, convex, and strongly monotone for every agent. If (c,y) is pareto efficient (where $c >> 0$), then there exists some e, θ, p such that (p,c,y) is a competitive equilibrium.*

3.4 Introducing Uncertainty

As I have pointed out in this section so far. We can interpret uncertainty (and time periods for that matter) as similar to adding different types of goods. It is actually not all that different. For example, a banana in a world in which a coin has flipped heads is different than a banana in a world in which a coin has flipped tails are different goods. The math doesn't care that these are qualitatively different states as long as we know that they are treated differently. I'm not going to formalize everything and just hope that it shows up in some of the exercises below.

3.5 Exercises

3.5.1 Prelim: Spring 2025

Consider an economy with 2 goods, grapes and wine, and two agents, Amy and Barney. Initially, there exists 1 grape. There exist two firms which can turn grapes into wine according to the production function $w = 2\sqrt{g}$, where g is the quantity of grapes put into the production function, and w is the amount of wine produced. Amy only likes grapes. Barney only likes wine. Each prefer more to less and each has an ability to consume any non-negative quantity of either good.

(a) What are the production and consumption sets?

> Solution: The production set is
>
> $$P = \{(-g, w) \in \mathbb{R}^2 : 0 \leq g \leq 1,\, 0 \leq w \leq 2\sqrt{g}\}.$$
>
> The consumption set, for each agent $i \in \{Amy, Barney\}$,
>
> $$C_i = \{(g, w) \in \mathbb{R}^2_+\}.$$
>
> ■

(b) Define an allocation given the environment described above. When is such an allocation feasible?

SOLUTION: Let $(g_A, w_A) = Z_A \in C_{Amy}$ and $(g_B, w_B) = Z_B \in C_{Barney}$ represent a particular consumption bundle for Amy and Barney respectively, and $(-g_1, w_1) = H_1 \in P$, $(-g_2, w_2) = H_2 \in P$ be elements of the production set for firm 1 and firm 2 respectively (given that they are both identical firms).

An **allocation** is the collection $\{Z_A, Z_B, H_1, H_2\}$. An allocation is **feasible** in this economy when, $g_A + g_B + g_1 + g_2 \leq 1$ and $w_A + w_B \leq w_1 + w_2$. ■

(c) Characterize the set of Pareto efficient allocations.

SOLUTION: Since each consumer prefers only different goods and grapes necessarily must be used to produce wine we can classify the set of pareto efficient allocations as

$$\boxed{\{(Z_A, Z_B, H_1, H_2) \,:\, g_B = 0,\, w_A = 0,\, g_1 = g_2,\, g_A + 2g_1 = 1,\, w_B = 4\sqrt{g_1})\}\,.}$$

■

(d) For general specifications of ownership of the initial grapes and the firms, define a competitive equilibrium. Will the allocation associated with any competitive equilibrium be efficient?

SOLUTION: A **competitive equilibrium** in this economy is a set of the collection of allocations $\{Z_A, Z_B, H_1, H_2\}$ and prices $\{p_g, p_w\}$ such that, given initial endowments of grapes e_A, e_B (where $e_A + e_B = 1$) and ownership of firm profits $\theta_{A1}, \theta_{A2}, \theta_{B1}, \theta_{B2}$ (where $\theta_{A1} + \theta_{B1} = \theta_{A2} + \theta_{B2} = 1$) such that

- $Z_A, Z_B \geq 0$.
- $p_g, p_w \in \mathbb{R}_+$
- (Grapes Market Clearing) $Z_A(g) + Z_B(g) \leq e_A + e_B + H_1(g) + H_2(g)$
- (Wine Market Clearing) $Z_A(w) + Z_B(w) \leq H_1(w) + H_2(w)$
- (Budget Constraint) $p_g Z_i(g) + p_w Z_i(w) \leq p_g e_i + \theta_{i1}(p_g H_1(g) + p_w H_1(w)) + \theta_{i2}(p_g H_2(g) + p_w H_2(w))$ for $i \in \{A, B\}$
- For all other agent allocations $\{\hat{Z}_A, \hat{Z}_B\}$ that are prefered to $\{Z_A, Z_B\}$, it must be that $\{\hat{Z}_A, \hat{Z}_B\}$ violates the budget constraint

- For all other firm allocations $\{\hat{H}_1, \hat{H}_2\}$, we have that it makes less profit for the firms.

Since this is a complete market, we can use the first welfare theorem and indeed conclude that any allocation associated with competitive equilibrium will be efficient. ■

(e) Find the set of competitive equilibria if Barney initially owns the grapes and Amy owns the firms?

SOLUTION: Let's first look at the firm's problem for each firm $i \in \{1, 2\}$.

$$\max_{g_i} \quad p_w(2\sqrt{g_i}) - p_g g_i$$

Taking the first order condition, we get that

$$\frac{p_w}{\sqrt{g_i}} - p_g = 0$$
$$\implies g_i = \left(\frac{p_w}{p_g}\right)^2$$

This then also yields profits $\pi_i = \frac{p_w^2}{p_g}$. With this, we can now consider both of Amy and Barney's maximization problems. For Barney, we know that he is initially endowed with all of the grapes and he only likes wine. This means that they make income p_g and uses all of that to buy wine at price p_w. Barney's consumption of wine is p_g/p_w (and he consumes 0 grapes).

For Amy, we know that her only source of income is through the firm of the profits and she like's to consume grapes at unit cost p_g. As such, we know that her consumption of grapes is $2\frac{p_w^2}{p_g^2}$ (and she consumes 0 units of wine).

Finally, to pin down the prices, we use the market clearing condition for grapes. We know that the sum of the grapes that Amy consumes and the grapes the firms use to produce wine must equal 1.

$$2\frac{p_w^2}{p_g^2} + 2\frac{p_w^2}{p_g^2} = 1,$$
$$\frac{p_w^2}{p_g^2} = \frac{1}{4},$$
$$\frac{p_w}{p_g} = \frac{1}{2},$$

$$2p_w = p_g.$$

Setting $p_w = 1$ as the numeraire, we get that $p_g = 2$. Putting everything together, we can finally classify the competitive equilibrium as

$$\boxed{Z_A = (0,2),\ Z_B = (1/2,0),\ H_1 = H_2 = (-1/4,1),\ p_g = 2,\ p_w = 1.}$$

We can get the whole set of equilibria by shifting the numeraire. ■

(f) Find the set of competitive equilibria if Amy initially owns the grapes and Barney owns the firms?

SOLUTION: A simple trap that people may fall into is conclude that Amy consumes all of her endowment of grapes simply because she prefers only grapes and has no reason to sell her endowment. This is not how competitive equilibrium works! The agents in the economy don't know that no one else has no endowment of grapes or any of their characteristics. Agents only interact with prices and the wealth generation from the value that prices imbues on their endowment. So even if this may end up being the case, the reason is not because Amy knows specific information about other people and refuses to share her endowment.

In this case, we actually get, however, that **there is no competitive equilibrium**.

To show this, note first that the firm's problem is the exact same as before. Now, we know that Barney get's the profits and consumes only wine, this means that he consumes $2\frac{p_w}{p_g}$ units of wine. For Amy, she sells her entire endowment of grapes at price p_g and uses it to buy only grapes, which means that she consumes $p_g/p_g = 1$ units of grapes.

To pin down prices, consider the market clearing condition for grapes

$$1 + 2\frac{p_w^2}{p_g^2} = 1,$$
$$\frac{p_w^2}{p_g^2} = 0.$$

Setting a numeraire of $p_g = 1$ we get that the only possible value for the price of wine is $p_w = 0$. This, however, is not possible because a price of 0 for wine would entail that Barney would buy an infinite amount of wine while firms are producing no wine (given that Amy consumes all of the grapes). This violates the market clearing for wine. As such, there is no possible competitive equilibrium in this market. ■

4 ECON 8103 (Rustichini)

This is the mini that I struggled with the most mostly because there is so much notation and pinning down intuition is very important as you go along or you will get very lost (as I did many a times).

4.1 Expected Utility Theory

Let's set up some primitive language to work with.

Definition 36 (Lotteries)**.** *Let C be some finite non-empty set of outcomes/consequences. A* ***simple lottery*** *over C is a probability measure*

$$p \in \Delta(C) := \left\{ p : C \to [0,1] \quad s.t. \quad \sum_{c \in C} p(c) = 1 \right\}.$$

For lotteries $p, q \in \Delta(C)$ and $\alpha \in [0,1]$, their ***mixture*** *is*

$$\alpha p + (1-\alpha) q \in \Delta(C), \quad (\alpha p + (1-\alpha)q)(c) = \alpha p(c) + (1-\alpha) q(c).$$

We denote $p\alpha q \equiv \alpha p + (1-\alpha)q$, and the way we interpret this is that we are randomizing between p and q with probability α and $1-\alpha$.

Rahman spends a little bit of time going through utility and expected utility representations again. We have done this in previous classes so I'm not going to go through them again, but let's talk about how agents may think about preferences over lotteries. We can use the classic preference relation $\succeq$, and discuss some properties that lottery preferences will have

1. **Completeness:** For any two lotteries $p, q \in \Delta(C)$, either $p \succeq q$ or $q \succeq p$, or both.

2. **Transitivity:** For all $p, q, r \in \Delta(C)$, if $p \succeq q$ and $q \succeq r$, then $p \succeq r$.

3. **Continuity:** For $p \succ q \succ r$, there exists some $\alpha, \beta \in (0,1)$ such that

$$\alpha p + (1-\alpha) r \succeq q \succeq \beta p + (1-\beta) r.$$

4. **Independence:** For all p, q, r and $\alpha \in (0,1)$:

$$p \succeq q \Leftrightarrow \alpha p + (1-\alpha) r \succeq \alpha q + (1-\alpha) r.$$

Completeness and transitivity are both very intuitive. Stuff we have seen before. Continuity says that "no outcome is infinitely better than another"; in other words, we can always find some lottery that sits in between two lotteries in terms of preferences. Independence is trying to relate some sense of a preference where mixing with another arbitrary "background" lottery preserves the rankings. These properties help give way to the following theorems

Theorem 4.1 (vNM Existence)**.** *A preference relation $\succeq$ admits an expected utility representation (EUR) if and only if it satisfies all four of the axioms above.*

Theorem 4.2 (vNM Uniqueness)**.** *This EUR from above is unique up to positive affine transformations: if u and v both represent $\succeq$, then $u(c) = Av(c) + B$ for some $A > 0, B \in \mathbb{R}$.*

4.2 Normal Form Games and Nash Equilibrium

With lotteries and utility representations in hand, let's start to set up the framework with which we will analyze games.

Definition 37 (Finite Normal Form Game)**.** *A **finite normal form game** is a tuple $(I, (A^i)_{i \in I}, (u^i)_{i \in I})$ where*

1. *$I = \{1, \ldots, n\}$ is a finite set of players.*
2. *A^i is a finite set of actions for player i. We also write $A = \prod_i A^i$ as the cardinal product of all the action sets for each player.*
3. *$u_i : A \to \mathbb{R}$ is player i's payoff function.*

In the game, players simultaneously choose actions without observing each other's choices.

*A **mixed strategy** for player i is some $s^i \in \Delta(A^i)$ (a vector of probabilities). The expected payoff is then*

$$u^i(s) = \sum_{a \in A} Pr_s(a) u^i(a) = \sum_{a \in A} \left(\prod_{j \in I} s^j(a^j) \right) u^i(a).$$

A pure strategy on the other hand is one where there is 100% certainty in one specific action. We can think of pure strategies as a special case of mixed strategies.

Having defined games and strategies, we can start thinking about equilibria concepts. Nash equilibrium is something that we have definitely seen in many other classes before but this is formalizing it with respect to the formalization of normal games.

Definition 38 (Nash Equilibrium)**.** *The **best response correspondence** for player i is*

$$BR^i(s^{-i}) = \left\{ r^i \in S^i \,:\, u^i(r^i, s^{-i}) \geq u^i(t^i, s^{-i}) \forall t^i \in S^i \right\}.$$

*A profile s^* is a **Nash Equilibrium** if $s^{*i} \in BR^i(s^{*-i})$ for every i. We can also write this as $s^* \in BR(s^*)$.*

Theorem 4.3 (Nash Theorem)**.** *Every finite normal game has at least one Nash equilibrium.*

This is a really cool result whose significance was lost on my when we were going through the proof in class. ANY GAME that we can represent as a finite normal form game is solved. This means *chess* has an "optimal" solution, *go* has an optimal solution, *shogi* has an optimal solution. Even though figuring out the equilibrium is an entirely different story, this result blew my mind.

4.3 Dominance, Iterated Elimination, and Rationalizability

We talked about what Nash equilibrium is, now let's talk about ways to find it. Generally, to give a rough structure, we can look for dominated strategies, check for pure nash equilibria, and then check for mixed nash equilibria (if a player mixes between two strategies then they must be indifferent between playing either of those pure strategies). Before we get into specific strategies, let's go back to thinking about best responses.

Definition 39 (Beliefs). *A **belief** of player i over A^{-i} is $\mu \in \Delta(A^{-i})$. Beliefs allow for correlation among opponents' actions. The expected payoff under belief μ is*

$$u^i(a^i, \mu) = \sum_{a^{-i} \in A^{-i}} \mu(a^{-i}) u^i(a^i, a^{-i}).$$

A mixed strategy s^i is a best response to μ if and only if it assigns probability 1 to the set of pure best responses:

$$s^i(BR^i_{A^i}(\mu)) = 1.$$

With this system of beliefs, we can get into dominance:

- a^i **strictly dominates** b^i (pure vs. pure) if $u^i(a^i, a^{-i}) > u^i(b^i, a^{-i})$ for all $a^{-i} \in A^{-i}$.
- a^i **weakly dominates** b^i (pure vs. pure) if $u^i(a^i, a^{-i}) \geq u^i(b^i, a^{-i})$ for all $a^{-i} \in A^{-i}$.
- a^{-i} is **strictly dominated by a mixed strategy** if $\exists s^i \in \Delta(A^i)$ such that $u^i(s^i, a^{-i}) > u^i(a^i, a^{-i})$ for all $a^{-i} \in A^{-i}$.
- a^{-i} is **weakly dominated by a mixed strategy** if $\exists s^i \in \Delta(A^i)$ such that $u^i(s^i, a^{-i}) \geq u^i(a^i, a^{-i})$ for all $a^{-i} \in A^{-i}$.
- a^i is a **never best response (NBR)** if there is no belief $\mu \in \Delta(A^{-i})$ making it a best response.

One way we can use these classifications to arrive is the **iterated elimination of striclty dominated strategies (IESDS)**. The concept behind this is quite simple and will be exemplified in a lot of the problems at the end of this section, but it involves narrowing down the action set of each individual until only a few actions and a nash equilibrium is left. Put into specific notation, IESDS is a sequence of trimmed action sets $(C^i_t)_{i \in I, t=0,1,\ldots}$ such that

1. $C^i_0 = A^i$ for all i.
2. $C^i_{t+1} \subseteq C^i_t$ for all i, t.
3. If $c^i \in C^i_t \setminus C^i_{t+1}$ then there exists some $s^i \in \Delta(C^i_t)$ such that

$$u^i(s^i, c^{-i}) > u^i(c^i, c^{-i}), \quad \forall c^{-i} \in C^{-i}_t.$$

A process of IESDS is complete if it is not possible to continue with further elimination of strategies. Critically, two properties of the terminal set C_T in this IESDS is that this outcome is both unique and Nash equilibrium always survives the elimination.

We can also do iterated elimination of weakly dominated strategies (IEWDS) in a similar manner as above, but critically although nash equilibria still survive the elimination process, we are not guaranteed that the terminal set is unique. One other way that we can try to think intuitively about this iterated dominance strategy is through beliefs. Specifically, the outcome of IESDS is a cet of actions that we consider to be "rationalizable", there exists some belief such that the actions inside this set weakly dominate the actions outside of this set.

Definition 40 (Rationalizability)**.** *A vector $(R^1, \ldots, R^n)$ with $R^i \subseteq A^i$ is **rationalizable** if for every i and every $a^i \in R_i$, there exists a belief $\mu \in \Delta(R^{-i})$ such that a^i is a best response*

$$u^i(a^i, \mu) \geq u^i(b^i, \mu)$$

for all $b^i \in A^i$.

The main result here is that if (C_T^i) is the outcome of a complete IESDS and (R^i) the maximal rationalizable vector, then $R^i = C_T^i$ for all i.

4.4 Extensive Form Games

Normal form games are a specific game style where agents all determine their action simultaneously. There are also games, however, where action takes place sequentially and we will build some infratructure for that now. Keep in mind, however, that we can also combine the two, where some sequential interaction leads to a normal form game with simultaneous action, example 4.8.1 is one form of this.

Definition 41 (Extensive Form Game)**.** *An **Extensive Form Game** is a collection of*

- *A set of players $I = \{1, \ldots, n\}$.*
- *A set of nodes $\{\alpha, X, Z\}$, α represents the inital node, X is the set of intermediate nodes, and Z is the set of terminal nodes.*
- *Player utilities $u^i : Z \to \mathbb{R}$ for all $i \in I$.*
- *A partial order $\succ$ on the nodes that orders the nodes. It is irreflexive, transitive, and "tree-like".*
- *A partition $P = \{P_1, \ldots, P^n\}$ of X. Each set. P^i, in the partition describes the set of nodes at which player i makes a move.*
- *For each $i \in I$, an information partition U^i of P^i. An information set groups nodes that a player cannot distinguish when it is their turn to move. A pure strategy must prescribe the same action at every node in the same information set – you can't condition on something you don't observe. If*

player 2 acts without knowing player 1's choice, all of player 1's successors are in one information set for player 2.

- *For each $i \in I$ and information set$u^i \in U^i$, a choice set C_{u^i}.*

We have discussed the setup, now let's talk about the type of strategies that one can commit themselves to in this game setup. The notation is a bit confusing here but the intuition is very simple. It is just a game tree where one action by an individual can branch out to several other game states. There are generally three types of ways that we can represent actions

- **Pure Strategies** are $s^i : U^i \to \bigcup_u C_u$ is a complete contingent plan specifying an action at every information set of player i, *including those never reach in equilibrium*. This part is trippy, here a complete action is a commitment to play a certain action if any specific information set is reached, it is not just a declaration of observed behavoir.

- **Behavioral Strategies** are β^i which specify an independent randomization at each information set. It says how likely we are to choose each of the actions available to the agent at each information set.

- **Mixed Strategies** simply just randomized over pure strategies. We can recover a behavioral strategy from mixed strategies and visa-versa as well.

4.5 Backward Induction and Subgame Perfect Equilibrium

With a different class of games and strategy concepts, comes a different class of equilibriua concepts.

Definition 42 (Perfect Information). *An extensive form game has **perfect information** if every information set I^i_j is a singleton. In other words, each node uniquely determines the history of play for each agent.*

On games of perfect information, we can do **backward induction** to solve for a nash equilibrium. There is a formal formulation of this in Aldo's notes, but I don't want to go through all that. Put simply, backward induction works recursively: at each decision node, the active player selects the action maximizing their continuation value. This yields a pure strategy profile and is the canonical solution for sequential games without information asymmetry.

Definition 43 (Subgame). *A **subgame**, $\Gamma(x)$, of some extensive form game is the subtree rooted at node x provided no information set is split: if an information set intersects the subtree, it is entirely contained within it.*

Definition 44 (Subgame Perfect Equilibrium). *A strategy profile β is a **subgame perfect equilibrium** (SPE) if its restriction to every subgame $\Gamma(x)$ is a nash equilibrium of $\Gamma(x)$. Every finite Extensive Form Game has at least one SPE!*

Theorem 4.4 (One-Stage Deviation Principle (OSDP))**.** *A strategy profile is SPE if and only if no player can gain by deviating at a single information set while conforming to the equilibrium everywhere else.*

4.6 Beliefs, Sequential Rationality, and Perfect Bayseian Equilibrium

Although backwards induction and SPEs eliminate non-credible behavoirs in subgames, they don't say anything about behavior at information sets that don't initiate further subgames. Perfect Bayesian Equilibrium (PBE) addresses this by requiring players to hold explicit *beliefs* about where they are in the game tree, and to act rationally given those beliefs. Before we talked about a perfect information world, but now we consider what happens in worlds without perfect information.

Definition 45 (Belief System)**.** *A* ***belief system*** *μ assigns a probability distribution $\mu(\cdot|I_j^i) \in \Delta(I_j^i)$ over nodes within each information set I_j^i*

An ***assesment*** *is a pair (β, μ) of a behavioral strategy profile and a belief system.*

Definition 46 (Perfect Bayesian Equilibrium)**.** *An assessment (β, μ) is a* ***Perfect Bayesian Equilibrium*** *(PBE) if:*

- *Sequential Rationality: At every information set, β maximizes each player's expected continuation payoff given μ and opponents' strategies.*
- *Bayesian Consistency: At every on-path information set (reached with positive probability), beliefs satisfy Bayes' rule:*
$$\mu(x|I_j^i) = \frac{PR_\beta(x)}{\sum_{y \in I_j^i} PR_\beta(y)}.$$
Off-path beliefs are unrestricted, subject only to sequential rationality.

The intuition here is that before we handled games where each agent knows exactly where they in the game at any given moment and so PBE adapts this by adding upon a set of beliefs that each agent may have and forces these beliefs to be consistent to count as a proper equilibrium. We can generally relate this to SPE as:

- Every SPE can be extended to a PBE (construct on-path beliefs via Bayes' rule; choose arbitrary off-path beliefs).
- PBE can further refine SPE by restricting off-path beliefs.
- PBE is weaker than Sequential Equilibrium, which additionally requires beliefs to be consistent limits of fully-mixed strategy sequences.

4.7 Correlated Equilibrium

Critically, nash equilibrium assumes that players randomize *independently*. Correlated, on the other hand, allows players to condition on a jointly observed signal (e.g. some mediatior's recommendation, a classic example are streetlights in a game where cars meet at an intersection). To formalize this, let's consider the classic normal form game that we have defined in prior sections.

Definition 47 (Correlated Equilibrium)**.** *A distribution $\mu \in \Delta(A)$ is a **correlated equilibrium** (CE) if for every player i and every recommendation action a^i with $\mu_{A^i}(a^i) > 0$, following the recommendation is optimal:*

$$\sum_{a^{-i} \in A^{-i}} \mu(a^i, a^{-i}) \left[u^i(a^i, a^{-i}) - u^i(b^i, a^{-i}) \right] \geq 0, \quad forall b^i \in A^i.$$

Equivalently: conditional on receiving recommendation a^i, player i weakly prefers following it over any deviation b^i.

In a NE, each player's randomization is independent – your mix carries no information about opponents' realizations. A correlated signal can carry conditional information: if I was told to Go, I know my partner was told to Stop. This conditional structure enables coordination that independent mixing cannot achieve.

4.8 Exercises

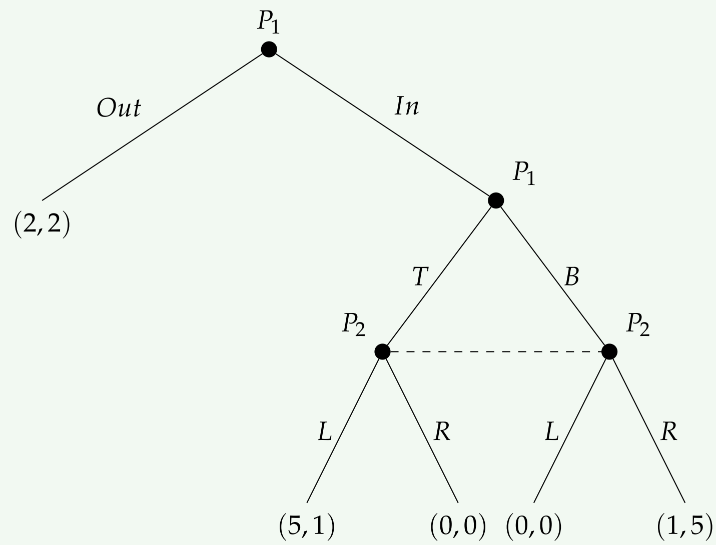
4.8.1 Prelim: Fall 2024

Consider the following game with two players. Player 1 moves first and chooses an action in the set $\{Out, In\}$. If he chooses Out the game is over and the two players receive a utility of $(2,2)$, with the first component paid to the first player. If Player 1 chooses In, then the two players play a matrix game called G, where each player has a two action set, $\{T, B\}$ for the first player and $\{L, R\}$ for the second. Utilities in G are:

	L	R
T	$5,1$	$0,0$
B	$0,0$	$1,5$

(1) Write the game in extensive form.

Solution: This is the game in extensive form:



■

(2) Write the game in normal form.

Solution: This is the game in normal form. Player 1's strategies are $\{OT, OB, IT, IB\}$ (e.g. OT means "play Out, and play T if In is reached") and Player 2's strategies are $\{L, R\}$:

	L	R
OT	$(2,2)$	$(2,2)$
OB	$(2,2)$	$(2,2)$
IT	$(5,1)$	$(0,0)$
IB	$(0,0)$	$(1,5)$

■

(3) Find all the Nash equilibria of the game.

Solution: Looking at the normal form game, the Nash equilibria are

$$\boxed{\{(OT, R),\ (OB, R),\ (IT, L)\}.}$$

	L	R
OT	$2,\mathbf{2}$	$\mathbf{2},\mathbf{2}$
OB	$2,\mathbf{2}$	$\mathbf{2},\mathbf{2}$
IT	$\mathbf{5},\mathbf{1}$	$0,0$
IB	$0,0$	$1,\mathbf{5}$

■

(4) Find all the subgame perfect equilibria of the game.

Solution: There are two subgames: the entire game itself, and the subgame that occurs if Player 1 chooses In. We already know the Nash Equilibria of the entire game, so let us consider the In subgame:

	L	R
T	$5,1$	$0,0$
B	$0,0$	$1,5$

The Nash equilibria of the subgame are $\{(T,L),\ (B,R)\}$. Matching these with the Nash Equilibria of the full game above, the subgame perfect equilibria of the game are

$$\boxed{\{(IT,L),\ (OB,R)\}.}$$

■

(5) Find the game resulting from iterated elimination of strictly dominated strategies.

Solution: Writing down the normal form game and crossing out strategies that are strictly dominated:

	L	R
OT	$2,2$	$2,2$
OB	$2,2$	$2,2$
IT	$5,1$	$0,0$
$\cancel{IB}$	$\cancel{0,0}$	$\cancel{1,5}$

IB is strictly dominated by OT (Player 1 always receives 2 from OT versus at most 1 from IB). No further strict dominations exist, so the resulting game is:

	L	R
OT	$2,2$	$2,2$
OB	$2,2$	$2,2$
IT	$5,1$	$0,0$

■

(6) Find the games resulting from iterated elimination of weakly dominated strategies.

SOLUTION: We proceed in the following steps:

(1) IT strictly dominates IB.

(2) L weakly dominates R (for Player 2).

(3) IT strictly dominates OT and OB.

	L	$\cancel{R}$
$\cancel{OT}$	$\cancel{2,2}$	$\cancel{2,2}$
$\cancel{OB}$	$\cancel{2,2}$	$\cancel{2,2}$
IT	$5,1$	$\cancel{0,0}$
$\cancel{IB}$	$\cancel{0,0}$	$\cancel{1,5}$

The resulting game is:

	L
IT	$5,1$

■

(7) Suppose that in a pre-play communication players have agreed to play the following strategy profile: "Player 1 plays Out. Player 2 will play R in the event that the game G is reached."

Suppose that in the actual game player 2 is actually called to move. What should player 2 think that player 1 just did, and what should player 2 choose? What does the iterated elimination of weakly dominated strategies suggest?

SOLUTION: Player 2 knows that by sticking to the agreed strategy, Player 1 would receive 2 units of utility. This means that Player 2 knows Player 1 would only deviate if they are intending to obtain more than 2 units of utility. As such, Player 2 assumes that Player 1 will play IT, and in order to obtain more than 0 units of utility, Player 2 will play L.

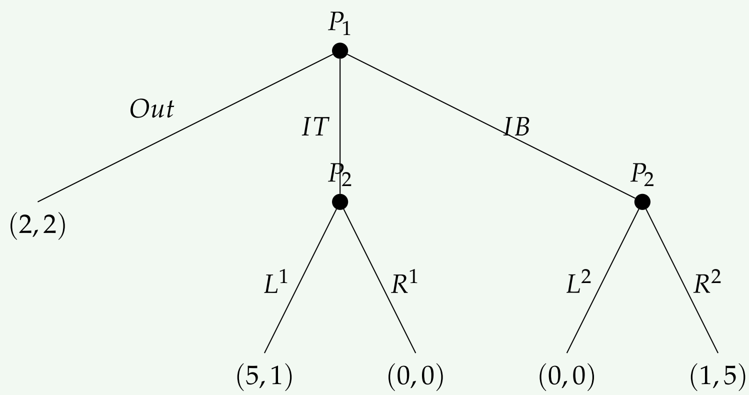
Iterated elimination of weakly dominated strategies also suggests the outcome

$$\boxed{(IT, L).}$$

■

(8) Consider the game with two players in which player 1 moves first and chooses an action in the set $\{Out, IT, IB\}$. If this player chooses Out then the game ends with payoffs $(2,2)$; if he chooses an action in $\{IT, IB\}$ then player 2 can choose an action in the set $\{L, R\}$ and payoffs are as in the game G with IT and IB replacing T and B respectively. Answer points 1–7.

SOLUTION: **(1)** Here is the extensive form game:



(2) Here is the normal form game. Player 2 now has four strategies $\{L^1L^2$, R^1L^2, L^1R^2, $R^1R^2\}$, where the superscript denotes the action in the IT branch (^{1}) and the IB branch (^{2}):

	L^1L^2	R^1L^2	L^1R^2	R^1R^2
OUT	$(2,2)$	$(2,2)$	$(2,2)$	$(2,2)$
IT	$(5,1)$	$(0,0)$	$(5,1)$	$(0,0)$
IB	$(0,0)$	$(0,0)$	$(1,5)$	$(1,5)$

(3) Starting from the normal form, the Nash equilibria are:

$$\{(IT,\ L^1L^2),\ (OUT,\ R^1L^2),\ (IT,\ L^1R^2),\ (OUT,\ R^1R^2)\}.$$

	L^1L^2	R^1L^2	L^1R^2	R^1R^2
OUT	2, **2**	**2**, **2**	2, **2**	**2**, **2**
IT	**5**, **1**	0, 0	**5**, **1**	0, 0
IB	0, 0	0, 0	1, **5**	1, **5**

(4) There are three subgames: the entire game, the IT subgame, and the IB subgame. In the IT subgame, L^1 is an equilibrium (Player 2 gets $1 > 0$). In the IB subgame, R^2 is an equilibrium (Player 2 gets $5 > 0$). Matching these with the Nash equilibria of the full game above, the SPE is:

$$\boxed{(IT,\ L^1,\ R^2).}$$

(5) We use iterated elimination of strictly dominated strategies. OUT strictly dominates IB (Player 1 always receives 2 from OUT versus at most 1 from IB). No further strict dominations remain.

	L^1L^2	R^1L^2	L^1R^2	R^1R^2
OUT	2, 2	2, 2	2, 2	2, 2
IT	5, 1	0, 0	5, 1	0, 0
~~IB~~	~~0, 0~~	~~0, 0~~	~~1, 5~~	~~1, 5~~

(6) We use iterated elimination of weakly dominated strategies:

- OUT strictly dominates $IB \Rightarrow$ eliminate IB.
- L^1L^2 weakly dominates R^1L^2 (for Player 2) $\Rightarrow$ eliminate R^1L^2.
- L^1L^2 weakly dominates R^1R^2 (for Player 2) $\Rightarrow$ eliminate R^1R^2.
- IT strictly dominates $OUT \Rightarrow$ eliminate OUT.

	L^1L^2	~~R^1L^2~~	L^1R^2	~~R^1R^2~~
~~OUT~~	~~2, 2~~	~~2, 2~~	~~2, 2~~	~~2, 2~~
IT	5, 1	~~0, 0~~	5, 1	~~0, 0~~
~~IB~~	~~0, 0~~	~~0, 0~~	~~1, 5~~	~~1, 5~~

The resulting game after elimination is:

	L^1L^2	L^1R^2
IT	$5,1$	$5,1$

(7) Again, we know that if Player 1 is deviating from OUT, they are trying to obtain more than 2 units of utility and will be choosing IT (since IB gives at most 1). As such, Player 2 will then choose L^1, giving either (L^1, L^2) or (L^1, R^2). ■

5 ECON 8104 (Rahman)

Personally, I feel like this mini is all over the place. There's a lot to learn and I frankly got lost a lot, similar to the last mini. To preface, there are five main categories that we discussed

1. Cooperative Theory
2. Social Choice
3. Mechanism Design
4. Information Economics
5. Dynamic games with and without information

Once again, I won't go too in depth into any of the proofs but I will state theorems and definitions and try to add some color commentary for intuition. Also, to be frank, the mechanism design section is AI generated.

5.1 Cooperative Theory

5.1.1 Two Person Bargaining Problem

Definition 48. *A **two person bargaining problem** is a pair (F, v), for $F \subseteq \mathbb{R}^2$ is closed and convex, $v \subseteq \mathbb{R}^2$ such that*

$$F_v = \{y \in \mathbb{R}^2 : y \geq v\} \cap F \neq \emptyset$$

*and bounded. (F, v) is called **essential** if $\exists y \in F$ such that $y >> v$.*

The way to interpret this is a two person game where F describes the set of all possible utilities $(u_1, u_2) \in F$ that agent 1 and agent 2 can get respectively by agreeing on some outcome through the game. v is the utility pair that will occur if there is disagreement. This is pretty abstracted from specific mechanics of a game and goes straight to the end utilities. Now consider the map $\varphi(F, v) \to \mathbb{R}^2$ which maps the bargaining problem to some outcome utilities (φ_1, φ_2), we call this the **bargaining solution**. There are a couple of axioms (characteristics of the bargaining problem and solution) that can help us pin down specific solutions

(i) **Efficiency:** For $\varphi(F,v) \in F$ and all $x \in F$, if $x \geq \varphi(F,v)$ then $x = \varphi(F,v)$.

(ii) **Individual Rationality:** $\varphi(F,v) \geq v$. Both people have to get at least v from the solution because either one can always opt out and achieve v.

(iii) **Scale Covariance:** For any $\lambda_1, \lambda_2 > 0$ and γ_1, γ_2. If

$$G := \{(\lambda_1 x_1 + \gamma_1, \lambda_2 x_2 + \gamma_2 \, : \, (x_1, x_2) \in F\}$$

and

$$w := (\lambda_1 v_1 + \gamma_1, \lambda_2 v_2 + \gamma_2),$$

then

$$\varphi(G,w) = (\lambda_1 \varphi_1(F,v) + \gamma_1, \lambda_2 \varphi_2(F,v) + \gamma_2).$$

(iv) **IIA:** For all closed and convex subsets $G \subseteq F$. If $\varphi(F,v) \in G$ then $\varphi(G,v) = \varphi(F,v)$.

(v) **Symmetry:** If $v_1 = v_2$ and $\{(x_2, x_1) \, : \, (x_1, x_2) \in F\} = F$ then $\varphi_1(F,v) = \varphi_2(F,v)$.

Theorem 5.1. *There exists a unique φ that satisfies axioms 1-5 and moreover it is the **Nash bargaining solution***

$$\varphi(F,v) \in \arg \max_{x \in F, x \geq v} (x_1 - v_1)(x_2 - v_2).$$

Example 1. *Consider an environment where agent one is risk neutral and has uility function $u_1(x) = x$ and agent two is risk averse and has utility function $u_2(x) = \sqrt{x}$. Consider the*

$$F = \{(y_1, y_2) \geq 0 : y_1 + y_2^2 \leq 1\}$$

and $v = (0,0)$. We know that the pareto frontier here is when $y_1 = 1 - y_2^2$ so using the theorem above, we need to maximize

$$\arg \max_{y \in F, x \geq 0} y_1 y_2 = \arg \max_{y_2 \in F, x \geq 0} (1 - y_2^2) y_2.$$

This then gives us

$$\varphi(F,v) = \left(\frac{2}{3}, \frac{1}{\sqrt{3}} \right).$$

Interestingly enough we can see that in bargaining situations like this, the risk-averse agent gets less utility in the Nash solution.

From our bargaining problem setup, there are two alternative solutions that we can consider as well

1. **Egalitarian solution:** Selects the pareto point $\varphi^E(F,v) = (x_1, x_2)$ from (F,v) that is efficient and also

$$x_1 - v_1 = x_2 - v_2.$$

2. **Utilitarian solution:** Selects the pareto point $\varphi^U(F,v) = (x_1, x_2)$ from (F,v) such that

$$x_1 + x_2 = \max_{y \in F} y_1 + y_2$$

Example 2. *Consider $v = 0$ and $F = \{(y_1, y_2) \geq 0 : y_1 + 2y_2 \leq 20\}$. The utilitarian maximizes $y_1 + y_2$ on F and we can see how in this case the result is $\varphi^U(F,v) = (20,0)$ whereas in a nash bargaining situation as above, the result would be $\varphi(F,v) = (10,5)$.*

5.1.2 Rational Threats

Disagreement payoffs don't need to be only some exogenously driven object. We can develop a specific mechanism through means of strategic "threat choices". Let each player i first choose a mixed strategy $\tau_i \in \Delta(A_i)$, if they fail to agree, then they play (τ_1, τ_2). We denote $v_i = u_i(\tau_1, \tau_2)$.

Definition 49. *A **rational threat equilibrium** is a pair (τ_1^*, τ_2^*) such that each τ_i^* is a best response to the Nash bargaining outcome induced by v, i.e.*

$$\tau_i^* \in \arg\max_{\tau_i} \quad \varphi_i^N \left(F, u_1(\tau_1, \tau_2^*), u_2(\tau_1, \tau_2^*)\right).$$

5.1.3 Alternating Offers

Consider the following situation, two players alternate offers over a pie of size 1. Player 1 proposes at odd dates, player 2 at even dates. After a rejection, with probability p_j the game ends in disagreement, giving continuation value v_j to player j.; with probability $1 - p_j$ the bargaining continues one more period (temporally discounted by $\delta_j \in (0,1)$). In other words, if y_2^* is player 2's continuation playoff after rejection, then to induce player 2 to accept, player 1 offers

$$x_2 = p_1 v_2 + (1 - p_1)\delta_2 y_2^*.$$

Similarly, the problem is mirrored for player 2 making an offer to player 1

$$x_1 = p_2 v_1 + (1 - p_2)\delta_1 y_1^*.$$

Since $y_2^* = 1 - x_1$, this leads to the unique SGPE payoff vector

$$x_2^* = \frac{p_1 v_2 + (1 - p_1)\delta_2[1 - p_2 v_1 - (1 - p_2)\delta_1]}{1 - (1 - p_1)\delta_2(1 - p_2)\delta_1}, \quad x_1^* = 1 - x_2^*.$$

5.1.4 Cooperative Games and the Core

Definition 50 (TU Cooperative Game)**.** *A TU Cooperative Game is a pair (N, v) with players $N = \{1, \ldots, n\}$ and characteristic function $v : 2^N \to \mathbb{R}$, $v(\emptyset) = 0$. Qualitatively, v is a mapping from any subset of players to some value and represents the total payof that a given coalition can guarantee itself* *regardless of what players outside of that coalition can do. For example, consider some situation with three firms who all are in a sector that experiences larger profits through collaboration. The respective payoffs could be*

$$v(\{f_1\}) = 0, v(\{f_1, f_2\}) = 3, v(\{f_1, f_2, f_3\}) = 9.$$

*We call v **superadditive** if $v(S \cup T) \geq v(S) + v(T)$ whenever $S \cap T = \emptyset$. We call v **monotone** if $S \subseteq T \implies v(S) \leq v(T)$.*

Definition 51 (The Core)**.** *An **imputation** is some $x \in \mathbb{R}^N$ with $\sum_i x_i = v(N)$ (efficiency) and $x_i \geq v(\{i\})$ (individually rational). Qualitatively it just represents a division of utility across the individuals in a coalition. As such, it must be that the division of utility achieves the total value possibly achieved in that coalition (efficiency). For individual rationality, this implies that an imputation must assign at least as much utility to a particular agent as is possible for them to achieve in a coalition by themselves.*

*The **core** is*

$$C(N, v) = \left\{ x \in \mathbb{R}^N : \sum_{i \in N} x_i = v(N), \sum_{i \in S} x_i \geq v(S) \quad \forall S \subseteq N \right\}.$$

Intuitively, the core represents the set of imputations where out of the grand coalition of all N agents, there is no smaller coalition that feels like it can do better by breaking away from the grand coalition.

As a simple example for the core, consider: three players and a "majority wins" game: $v(S) = 1$ if $|S| \geq 2$, and $v(S) = 0$ otherwise. The core is empty here. Any proposed division (x_1, x_2, x_3) with $x_1 + x_2 + x_3 = 1$ will always leave some two-player coalition getting less than 1 collectively, since you can't give every pair a combined share of 1 simultaneously. So some coalition always wants to defect.

We can extend this idea of a core to stuff we are familiar with such as walrasian equilibrium. Consider an exchange economy with N agents, ℓ goods plus money, endowments $\omega_i \in \mathbb{R}^\ell_+$, and quasilinear utility $v_i(z_i) + m_i$. A walrasian equilibrium (as we are definitely familiar with by now) is a (x^*, p^*) with $p^* \in \mathbb{R}^\ell_+$ such that the allocations satisfy the individual optimization problems of each agent given prices and the allocations also satisfy market clearing. The corresponding coalition value function of the exchange economy is

$$v(S) = \max \left\{ \sum_{i \in S} v_i(x_i) : \sum_{i \in S} x_i = \sum_{i \in S} \omega_i,\, x_i \geq 0 \right\}.$$

5.2 Social Choice

A significant portion of the class was dedicated to Arrow's theorem and Gibbard-Satterthwaite and their corresponding proofs. These are definitely important concepts, but the proofs themselves are unlikely to show up in any exam so I will abstain from diving in too deep.

Consider the following environment.

- Finite set of individuals $N = \{1, \ldots, N\}$.

- Finite set of alternatives A such that $|A| > 3$.
- Each agent has compete and transitive preference R_i

Definition 52. *A* ***social welfare function*** $f : \mathcal{R}^N \to \mathcal{R}$*, where* $\mathcal{R}$ *are rational preferences on* A*, is a function that maps individual agent preferences to a social choice considering everyone's preferences. Here are some properties that a social welfare function could have.*

- ***Universality (UD)****: the domain is* $\mathcal{R}^N$*.*
- ***Unanimity****: If* aP_ib *for all* i*, then* aPb*.*
- ***Independence of Irrelevant Alternatives (IIA)****: If some* aR_ib *implies* $aR_i'b$ *for any* i*, then* $af(R)b \implies af(R')b$*.*
- ***Dictatorial****: There exists exactly one agent* i *such that* $aP_ib \implies aPb$*.*

Theorem 5.2 (Arrow's Impossibility Theorem (1951))**.** *If* $|A| > 3$ *and* N *is finite, then every UD + Unanimous + IIA social welfare function is dictatorial.*

Definition 53. *A* ***social choice function*** *on the other hand doesn't represent a map between individual to societal preferences but a map between linear ordering (preferences) and a single societal choice. Here are some properties that a social choice function could have.*

- ***Strategy-Proof****:* $f(P_i, P_{-i})R_if(P_i', P_{-i})$ *for every* i, P_i, P_i', P_{-i}*.*
- ***Monotone****: Whenever the script upper contour set of* $f(P)$ *weakly shrinks in each* $P_i \to P_i'$*, then* $f(P) = f(P')$*.*
- ***Dictatorial****: There exists one* i *such that* $f(P)$ *is top ranked by* P_i *always.*

Theorem 5.3 (Gibbard-Satterhwaite)**.** *If* $|A| \geq 3$*,* f *is onto, and SP with universal domain, then* f *is dictatorial.*

5.3 Mechanism Design

I think this is the chonkiest section out of all of them. From my perfunctory glance of the prelim questions, it looks like this section is related to most of the prelim questions as well.

5.3.1 Vickery-Clarke-Groves (VGC) Mechanisms

Here is the **basic situation**. There is a group of agents who need to make a collective decision q — for example whether to build a public good, or who should receive an object. Each agent i has a private type t_i that determines how they value different outcomes. The big "social designer" wants to implement a "good" outcome but faces the problem that types are private information — agents might lie about their preferences to manipulate the decision in their favor. Critically, agent's utilities are all quasi-linear which means that the marginal utility of money is one and there are no constraints and the amount of money. To recap, there are:

- Agents, $N = \{1, \ldots, n\}$.
- Set of non-money allocations and money payments,

$$A := \left\{ (p, q) \,:\, q \in Q, p_i \in \mathbb{R}\, \forall i \in N, \sum_i p_i \geq 0 \right\}.$$

- Non-money allocations q.
- Money payments by i to the designer p_i
- Utility functions

$$u_i(p, q, t_i) = v_i(q, t_i) - p_i.$$

- Types $t_i \in T_i$ with $T = \Pi_{i=1}^n T_i$, and $t = (t_1, \ldots, t_n) = (t_i, t_{-i})$

The designer wants to implement the *ex-post efficient outcome* — the q^* that maximizes total surplus:

$$q^*(t) \in \arg\max_q \sum_i v_i(q, t_i) \tag{3}$$

The question is whether the designer can get agents to truthfully reveal their types so that q^* can actually be computed and implemented.

Just asking agents to report their types does not work. Suppose the decision is whether to build a public good that costs c. Agent i knows that if they report a higher value, the good is more likely to be built. If they want the good built, they have an incentive to overreport. If they do not want it built, they have an incentive to underreport. Truthful revelation is not generally incentive compatible.

Let's come back now to social choice functions, in this environment, a SCF takes on the form

$$t \xrightarrow{f} (p_1(t), \ldots, p_n(t), q(t))$$

such that $\sum_i p_i(t) \geq 0$ for all $t \in T$.

Definition 54 (Ex-Post Efficient). *f is ex-post efficient (EPE) in non-money if*

$$\sum_{i=1}^n v_i(q(t), t_i) \geq \sum_{i=1}^n v_i(q', t_i) \quad \forall t \in T, q' \in Q.$$

Definition 55 (Strategy Proof). *f is SP if for all (i, t_i, s_i, t_{-i}), we have*

$$v_i(q(t_i, t_{-i}), t_i) - p_i(t_i, t_{-i}) \geq v_i(q(s_i, t_{-i}), t_i) - p_i(s_i, t_{-i}).$$

Theorem 5.4 (Groves, 1973). *Let $f = (p, q^*)$ be an EPE social choice function. If $\forall i \in N$, there exists* *some h_i such that:*

$$-p_i(t) = \sum_{j \neq i} v_j(q^*(t), t)j) - h_i(t_{-i}), \quad \forall t \in T$$

Then f is strategy proof.

Groves' theorem says that the following transfer scheme makes truthful reporting a *dominant strategy* — meaning it is optimal for agent i to report truthfully regardless of what everyone else reports:

$$p_i(t) = h_i(t_{-i}) - \sum_{j \neq i} v_j(q^*(t), t_j) \tag{4}$$

where h_i is any function that depends only on other agents' reports t_{-i}, not on i's own report. With Groves transfers, agent i's total payoff from reporting t_i' is:

$$v_i(q^*(t_i', t_{-i}), t_i) + \sum_{j \neq i} v_j(q^*(t_i', t_{-i}), t_j) - h_i(t_{-i}) \tag{5}$$

The last term $h_i(t_{-i})$ does not depend on i's report so it is irrelevant to i's decision. The first two terms together are just:

$$\sum_j v_j(q^*(t_i', t_{-i}), t_j) \tag{6}$$

the *total social surplus* given the outcome induced by i's report. This is maximized when q^* is the efficient outcome, which happens exactly when i reports truthfully. So agent i is effectively choosing the outcome to maximize total social surplus — their private incentives are perfectly aligned with social efficiency. The Clarke mechanism is the most natural special case of Groves, where:

$$h_i(t_{-i}) = \sum_{j \neq i} v_j(q^*_{-i}(t_{-i}), t_j) \tag{7}$$

and q^*_{-i} is the efficient outcome *without agent i*. This gives transfers:

$$p_i(t) = \sum_{j \neq i} \left[v_j(q^*_{-i}(t_{-i}), t_j) - v_j(q^*(t), t_j) \right] \tag{8}$$

This is the *externality agent i imposes on everyone else* by participating. If agent i's presence changes the decision — pivoting the outcome from q^*_{-i} to q^* — then i pays for the harm they cause to others. If i's presence does not change the decision, they pay nothing.

The cleanest example is a single object auction where agent i values the object at t_i. The efficient outcome is to give the object to the highest type. Without agent i, the object goes to the highest type among everyone else. With agent i present, if i wins, the outcome changes — i has pivoted the decision. The externality i imposes is depriving the second highest bidder of the object, so i pays the second highest value. This is exactly the *Vickrey second price auction* — you win by having the highest value but pay the second highest bid. Losers do not change the outcome so they pay nothing.

The Clarke mechanism collects payments $p_i \geq 0$ from agents who are pivotal — it runs a *surplus*. This surplus cannot be returned to the agents without destroying incentive compatibility. So the mechanism generically *fails budget balance*.

The Green–Laffont / Holmström theorem makes this precise — there is *no mechanism* that simultaneously achieves:

- Dominant strategy incentive compatibility
- Ex-post efficiency
- Budget balance

The proof goes through Groves' theorem. Any DSIC and efficient mechanism must be a Groves scheme. Budget balance requires $\sum_i p_i = 0$, which means:

$$\sum_i h_i(t_{-i}) = (n-1)W(t) \tag{9}$$

where $W(t) = \sum_j v_j(q^*(t), t_j)$ is total surplus. The left side depends on t only through t_{-i} for each i, while the right side depends on all of t through cross terms. One can show this equality is generically impossible — the functions do not have the required separability structure.

The VCG family of mechanisms gets self-interested agents to truthfully reveal private information by making each agent the residual claimant of total social surplus. The cost is that the budget cannot simultaneously be closed. This tradeoff between efficiency, incentive compatibility, and budget balance is one of the central impossibility results of mechanism design, and everything downstream — Myerson auctions, Mussa–Rosen screening — is grappling with variations of this same tension.

5.3.2 Optimal Auction Design

A seller has a single object to sell. There are n bidders, each with a private value t_i for the object drawn independently from some distribution F_i with density f_i on $[a_i, b_i]$. The seller wants to design an auction — a rule for who gets the object and how much everyone pays — to maximize expected revenue. The question is what the optimal such rule looks like. By the revelation principle, without loss of generality the seller can restrict attention to *direct mechanisms* where each bidder simply reports their type t_i and the mechanism maps reports to an allocation $q(t)$ and payments $p(t)$. A direct mechanism (q, p) consists of:

- $q_i(t)$ — the probability bidder i gets the object given reports t
- $p_i(t)$ — the payment bidder i makes

The mechanism must satisfy two constraints for each bidder i. Define bidder i's interim expected utility at type t_i as:

$$U_i(t_i) = E_{t_{-i}}\left[t_i q_i(t_i, t_{-i}) - p_i(t_i, t_{-i})\right] \tag{10}$$

and the interim probability of winning as $Q_i(t_i) = E_{t_{-i}}[q_i(t_i, t_{-i})]$.

Incentive compatibility (IC): each bidder prefers to report truthfully:

$$U_i(t_i) \geq E_{t_{-i}}\left[t_i q_i(t_i', t_{-i}) - p_i(t_i', t_{-i})\right] \quad \forall t_i, t_i' \tag{11}$$

Individual rationality (IR): each bidder prefers to participate:

$$U_i(t_i) \geq 0 \quad \forall t_i \tag{12}$$

Applying the envelope theorem to the IC constraint characterizes what incentive compatibility implies. IC requires two things. First, Q_i must be *nondecreasing* — higher types must win more often. Second, the envelope condition pins down surplus:

$$U_i(t_i) = U_i(a_i) + \int_{a_i}^{t_i} Q_i(s)\, ds \tag{13}$$

So once the allocation rule q and the lowest type's surplus $U_i(a_i)$ are known, everything else is determined. IR pins down $U_i(a_i) = 0$ at the optimum — the lowest type gets zero surplus. Substituting the envelope condition into the revenue expression and integrating by parts gives:

$$E[\text{revenue}] = \sum_i E\left[q_i(t) \cdot \psi_i(t_i)\right] \tag{14}$$

where $\psi_i(t_i)$ is the *virtual valuation*:

$$\psi_i(t_i) = t_i - \frac{1 - F_i(t_i)}{f_i(t_i)} \tag{15}$$

The virtual valuation has two components. The first term t_i is the bidder's actual value. The second term $(1 - F_i(t_i))/f_i(t_i)$ is the *information rent* — the surplus the seller must give up to type t_i to prevent lower types from mimicking them. In the *regular case* where ψ_i is strictly increasing for each i, the revenue maximizing auction:

- Gives the object to the bidder with the highest virtual valuation, provided $\max_i \psi_i(t_i) \geq 0$
- Does not sell if all virtual valuations are negative
- Charges each bidder according to the envelope formula

The reserve price emerges naturally — do not sell to a bidder whose virtual valuation is negative, even if they are the highest bidder. This means the seller sometimes inefficiently withholds the object, which is the cost of revenue maximization. If $t_i \sim U[0,1]$ i.i.d., then $\psi(t) = 2t - 1$. Virtual valuation is nonnegative iff $t \geq 1/2$. The optimal auction is a *second price auction with reserve price* $r^* = 1/2$ — give the object to the highest bidder provided they exceed the reserve, and charge them the maximum of the second highest bid and the reserve. With $n = 2$ bidders this generates expected revenue 5/12 versus 1/3 without a reserve. When bidders have different distributions F_i, their virtual valuation functions ψ_i differ. The optimal auction gives the object to the bidder with the highest virtual valuation — which need not be the bidder with the highest actual value. This generates a bias in favor of weaker bidders: a bidder from a distribution with lower values might have a higher virtual valuation and win even against a stronger bidder with a higher true value, because inducing competition among asymmetric bidders benefits the seller. When ψ_i is not monotone — the non-regular case — the simple pointwise maximization breaks down because it would generate a non-monotone allocation rule, violating IC. Myerson's solution is *ironing*: replace ψ_i with its ironed version $\bar{\psi}_i$ by concavifying the integrated virtual valuation:

$$H_i(t) = \int_{a_i}^{t} \psi_i(s) f_i(s)\, ds \tag{16}$$

On intervals where H_i is concavified, $\bar{\psi}_i$ is flat — meaning the allocation is randomized uniformly over those types. This restores monotonicity of the allocation rule while preserving optimality. Any two IC mechanisms with the same allocation rule q and the same lowest type surplus $U_i(a_i)$ yield the same expected revenue. In particular, first price and second price auctions with symmetric bidders generate identical expected revenue despite looking very different — in the first price auction bidders shade their bids, but the shading exactly offsets the higher winning payment, leaving revenue unchanged.

5.3.3 Monopolistic Screening

A monopolist sells a good that comes in varying qualities q to a population of consumers who differ in how much they value quality. Each consumer has a private type θ — their willingness to pay for quality — and the monopolist cannot observe θ directly. The monopolist designs a *menu of contracts* (q, T) — quality-price pairs — and lets consumers self-select into the contract intended for their type. Start with two types $\underline{\theta} < \bar{\theta}$ with probabilities $\underline{p}$ and $\bar{p}$. Consumer utility is:

$$u(q, T; \theta) = \theta v(q) - T \tag{17}$$

where v is increasing and concave. The monopolist's cost of producing quality q is cq. The monopolist maximizes profit subject to four constraints:

$$\underline{\theta} v(\underline{q}) - \underline{T} \geq 0 \tag{$\mathrm{IR}_{\underline{\theta}}$}$$

$$\bar{\theta} v(\bar{q}) - \bar{T} \geq 0 \tag{$\mathrm{IR}_{\bar{\theta}}$}$$

$$\underline{\theta} v(\underline{q}) - \underline{T} \geq \underline{\theta} v(\bar{q}) - \bar{T} \tag{$\mathrm{IC}_{\underline{\theta}}$}$$

$$\bar{\theta} v(\bar{q}) - \bar{T} \geq \bar{\theta} v(\underline{q}) - \underline{T} \tag{$\mathrm{IC}_{\bar{\theta}}$}$$

At the optimum:

- **Low type's IR binds:** $\underline{T} = \underline{\theta} v(\underline{q})$. The monopolist extracts all surplus from the low type.

- **High type's downward IC binds:** $\bar{\theta}v(\bar{q}) - \bar{T} = \bar{\theta}v(\underline{q}) - \underline{T}$. The high type is exactly indifferent between their contract and the low type's contract.
- **High type's IR is slack:** the high type receives positive *information rent* equal to $(\bar{\theta} - \underline{\theta})v(\underline{q}) \geq 0$.
- **Low type's upward IC is slack:** the low type never wants to mimic the high type.

Substituting the binding constraints and optimizing yields:

$$\bar{\theta}v'(\bar{q}) = c \qquad \text{(no distortion at top)}$$

$$\underline{\theta}v'(\underline{q}) = c + \frac{\bar{p}}{\underline{p}}(\bar{\theta} - \underline{\theta})v'(\underline{q}) \qquad \text{(downward distortion at bottom)}$$

The high type gets the efficient quantity. The low type gets less than the first best quantity — the monopolist distorts quality downward to make the low type contract less attractive to the high type, thereby reducing the information rent paid to the high type. Sometimes it is optimal for the monopolist to exclude the low type entirely, setting $\underline{q} = 0$. This happens when the low type population is small relative to the high type population, so the monopolist finds it more profitable to focus entirely on extracting from the high type. With a continuum of types $\theta \sim F$ on $[\underline{\theta}, \bar{\theta}]$ with density f and single crossing condition $\partial^2 v / \partial q \partial \theta > 0$, the IC constraint reduces to two conditions. First, $q(\theta)$ must be *weakly increasing*. Second, the envelope condition:

$$U'(\theta) = v_\theta(q(\theta), \theta) \tag{18}$$

pins down the consumer's surplus path given the allocation rule. Substituting the envelope condition and integrating by parts transforms the monopolist's profit into a *virtual surplus* maximization. The optimal quantity for each type solves:

$$q^*(\theta) = \arg\max_q \left[v(q, \theta) - \frac{1 - F(\theta)}{f(\theta)} v_\theta(q, \theta) - cq \right] \tag{19}$$

The term $\frac{1-F(\theta)}{f(\theta)} v_\theta(q, \theta)$ is the *information rent derivative* — it captures the cost to the monopolist of giving type θ more quality, because all types above θ must also receive more surplus to maintain IC. At the highest type $\bar{\theta}$, the information rent term vanishes since $F(\bar{\theta}) = 1$:

$$\frac{1 - F(\bar{\theta})}{f(\bar{\theta})} = 0 \tag{20}$$

So the virtual surplus condition collapses to the first best condition — the top type always gets efficient quality regardless of the distribution or utility function. If the pointwise optimal $q^*(\theta)$ is not increasing, the solution requires *ironing* — replacing the virtual surplus with its concavified version on the non-monotone region. This corresponds to *bunching*: a positive mass of types all receive the same quality level. Bunching is the screening analogue of Myerson's ironing and arises for similar reasons. The mathematical structure is identical to Myerson's optimal auction:

- Both use the envelope theorem to reduce the IC constraint to a differential equation
- Both use integration by parts to write the objective as a virtual surplus
- Both feature no distortion at the top and downward distortion elsewhere

The difference is economic context — Myerson allocates a fixed object across competing bidders, while Mussa–Rosen designs a quality-price menu for a single consumer with private willingness to pay.

5.4 Information Economics

This section will cover a selection of a handful of different types of problems that revolve around the idea of information.

5.4.1 Akerlof's Lemons

Consider an environment with a buyer and a seller in a car market. The seller has a car that could either be a new car or a lemon (old beat down car) which both cost $\overline{c}$ and $\underline{c}$ respectively. The buyer also assigns value $\overline{v}$ and $\underline{v}$ to the new car or lemon respectively. The seller knows what time of car they are trying to sell and therefore both v and c, while the buyer does not know until they have finished buying the car.

The seller faces cost c which is unform over $\{\underline{c}, \overline{c}\}$. The buyers value is v and is also uniformly over $\{\underline{v}, \overline{v}\}$. We also have that $\overline{v} > \overline{c} > \underline{v} > \underline{c}$ and crucially $\frac{\underline{v}+\overline{v}}{2} < \overline{c}$. In this game, the buyer does not know his own value and the seller knows both. Given price P, there are several ways the game can shape out.

- If $P < \underline{c}$ then there is no trade because the seller would never make money
- If $\underline{c} \leq P < \overline{c}$, then the seller would offer if the car was a low quality one. The buyer's expected value conditional on the seller making an offer in the first place (which only happens when the car is a lemon) is $\underline{v}$. This means that a transaction only successfuly occurs if $P \leq \underline{v}$
- If $P \geq \overline{c}$, then the seller would offer a trade in any realization for the quality of the car. Since the buyer doesn't know the quality of the car, they only buy if their expected value is greater than the price. We know from the setup, however, that $\frac{\underline{v}+\overline{v}}{2} < \overline{c}$ and as such $\frac{\underline{v}+\overline{v}}{2} < P$. This means that a trade does not take place.

The main takeaway is that the assymetric nature of information leads to the only trades that take place being ones that involve lemons even if there is a situation where there is a high quality car that the buyer would be willing to purchase had they known it was a high quality car.

5.4.2 Rothschild-Stigliz Insurance

Consider an environment with two types of people, high risk (H) and low risk (L), where high risk involves a higher probability of an accident ($p_H vs p_L$). An insurance company wants to offer contracts but cannot observe who is high or low risk. Each person knows their own type. A contract specifies two wealth levels W_1 and W_2 where W_1 is the wealth if no accident occurs and W_2 is the wealth if an accident occurs. Full insurance means $W_1 = W_2$ (wealth is the same in either situation), and partial insurance means that $W_2 < W_1$.

Under full information, the insurer would know who is who and offer each type a personalized contract based on their risk of accident but under assymetric information where the insurer does not know who is who, the high risk individuals would always opt for the contract offered to low risk individuals since they have to pay less for similar coverage. To cope with this, the insurer has to offer contracts in a manner such that each individual will self select into their own group.

The way to do this is to under-insure the low risk type. By offering low risk people only partial insurance, you make that contract less attractive to high risk people, who value full insurance more because they face accidents more often. High risk people still get full insurance on their own zero profit line Z_H.

Theorem 5.5 (Rothschild-Stiglitz)**.** *A separating equilibrium (α_L, α_H) exists if and only if the proportion of H-types λ exceeds a threshold $\overline{\lambda}$ (otherwise a pooling contract would attract both types and break the separating equilibrium). If $\lambda < \overline{\lambda}$, no equilibrium exists in pure contracts.*

5.4.3 Signaling (Leland-Pyle 1977)

Consider an environment where an entrepreneur has a project that requires capital K with random future returns $\tilde{X}$ where $\mathbb{E}[\tilde{X}] = \mu$. Crucially, μ is the entrepreneur's private information and the overall market does not know this information. This is the classic lemons problem applied to financial markets. If the market can't distinguish good from bad projects, it prices everything at some average value. Good entrepreneurs are undervalued and bad entrepreneurs are overvalued. In the worst case, good projects don't get funded at all because entrepreneurs can't credibly communicate quality.

To get past this, Leland and Pyle's insight is that the entrepreneur can signal quality through how much equity they retain in their own project. The idea is: A good entrepreneur (high μ) is willing to hold a lot of their own equity because they know the project is valuable — the expected return justifies bearing the idiosyncratic risk of holding a concentrated position. A bad entrepreneur (μ) finds it too costly to mimic this. If they retained a lot of equity in a bad project, they'd be bearing a lot of idiosyncratic risk for a project they know isn't worth much. The cost of holding concentrated risk outweighs any benefit from fooling the market. So equity retention is a credible signal precisely because it is costly, and the cost is asymmetric across types - it hurts bad entrepreneurs more than good ones.

Formally, the entrepreneur has initial wealth W_0 and chooses a retained equity share $\alpha \in [0,1]$, raises $(1-\alpha)V(\mu)$ from outside investors, debt D, and market portfolio $\tilde{M}$ position β. The market observes α and prices the project accordingly via a price schedule $V(\mu)$. The entrepreneur has mean-variance preferences:

$$U = E[\tilde{W}] - \frac{b}{2}\text{Var}(\tilde{W}) \tag{21}$$

which corresponds to CARA utility with risk aversion coefficient b.

Taking $V(\mu)$ as given, the entrepreneur chooses α, β, D to maximize U. The FOC with respect to β gives the standard CAPM demand for the market portfolio. The FOC with respect to α gives:

$$\frac{\partial U}{\partial \alpha} = \mu - r_f V(\mu) - b\left[\alpha \text{Var}(\tilde{X}) + \beta \text{Cov}(\tilde{X}, \tilde{M})\right] + \alpha V'(\mu)\frac{\partial \mu}{\partial \alpha} = 0 \tag{22}$$

The market rationality condition requires the market to correctly infer μ from α, so $V(\mu)$ must satisfy:

$$V(\mu) = \frac{\mu - b\alpha \text{Cov}(\tilde{X}, \tilde{M})}{1 + r_f} \tag{23}$$

which is the CAPM pricing formula — the project is worth its expected return minus a risk premium for the systematic component of its risk.

For a separating equilibrium, we need $\alpha(\mu)$ to be strictly increasing. The single crossing condition requires that the marginal cost of retaining equity is lower for high μ types. The slope of the entrepreneur's indifference curves in (α, V) space is:

$$\left.\frac{dV}{d\alpha}\right|_{\bar{U}} = \frac{b\alpha \text{Var}(\tilde{X}) + b\beta \text{Cov}(\tilde{X}, \tilde{M}) - \mu + r_f V}{r_f \alpha} \tag{24}$$

This slope is *decreasing in* μ — a higher μ entrepreneur requires less additional valuation V to compensate for holding more equity α. This is the single crossing property that makes separation possible.

In a separating equilibrium the market's belief must be consistent, so $V = V(\mu(\alpha))$. Differentiating the market pricing equation with respect to α and combining with the entrepreneur's FOC gives a first-order ODE in $V(\mu)$:

$$V'(\mu) = \frac{1 - b\alpha'(\mu)\text{Cov}(\tilde{X}, \tilde{M})}{1 + r_f} \tag{25}$$

with boundary condition $V(\underline{\mu}) = \underline{V}$ at the lowest type, who has no incentive to signal and receives the full information price. This ODE pins down both $V(\mu)$ and $\alpha(\mu)$ simultaneously.

Under full information the entrepreneur would choose α^{FI} to minimize risk bearing:

$$\alpha^{FI} = -\frac{\text{Cov}(\tilde{X}, \tilde{M})}{\text{Var}(\tilde{X})} \tag{26}$$

which is the hedge ratio — the share that minimizes exposure to idiosyncratic risk. In the signaling equilibrium the entrepreneur holds $\alpha^*(\mu) > \alpha^{FI}$, strictly more than the efficient amount. The deadweight cost of signaling in utils is:

$$\Delta U = \frac{b}{2}\left(\alpha^*(\mu) - \alpha^{FI}\right)^2 \mathrm{Var}(\tilde{X}) \tag{27}$$

This is increasing in risk aversion b, the variance of the project $\mathrm{Var}(\tilde{X})$, and the gap between the signaling and efficient equity shares — the more risk averse the entrepreneur and the riskier the project, the more painful it is to hold the excess equity required to signal quality.

5.4.4 Cheap Talk (Crawford-Sobel)

Let's first setup the problem, there is a sender who has private information about the state of the world $\omega \sim U[0,1]$, and a reciever who takes and action $x \in \mathbb{R}$ based on whatever message the sender sends. The sender and receiver have different preferences over what action gets taken — specifically the sender wants the action to be a bit higher than the receiver does, captured by the bias $b > 0$. In Leland-Pyle the signal was costly — the entrepreneur had to actually hold equity, which was painful. Here the sender can send any message at zero cost. There is no commitment device, no costly action, nothing stopping the sender from lying. This is what "cheap talk" means — the message itself is costless and non-binding.

Utilities are $u_S = -(x-\omega-b)^2$ and $u_R = -(x-\omega)^2$ for senders and recievers respectively. Clearly the sender wants $w = \omega + b$ and the reciever wants $x = \omega$. The key difference here is that if messages are costless then why would the receiver ever believe them, and if the receiver never believes the messages then what incentive does the sender have to send truthful messages? The equilibrium in which no neither agent believes the other is called the **babbling equilibrium** where the receiver ignores all messages and just plays

$$x = \mathbb{E}[\omega] = 1/2.$$

On the other hand, full relevation of always telling the truth also fails the sender since the receiver will always set $w = \omega$ and the sender wants them to set it to something higher. As such, the sender has an incentive to send some message ω' such that $\omega' > \omega$.

Crawford and Sobel's insight is that partial information transmission is possible through a partition equilibrium. The idea is: Instead of reporting the exact value of ω, the sender credibly communicates only which interval ω falls into. The receiver then plays the midpoint of that interval. The sender is willing to be truthful about which interval they're in — but not about the exact value within the interval. Formally the state space $[0,1]$ is divided into N intervals $[0,\omega_1], [\omega_1,\omega_2], \ldots, [\omega_{N-1}, 1]$. The sender reports which interval ω falls in, and the receiver plays the midpoint $\overline{x}_i$ of the reported interval.

This works because at each boundary point ω_i, the sender must be indifferent between reporting the lower interval and the upper interval. If they striclty preferred one side, the equilibrium would inravel. This indifferences conditions is

$$-(\overline{x}_i - \omega_i - b)^2 = -(\bar{x}_{i+1} - \omega_i - b)^2,$$

which says the sender at ω_i is equally happy with action $\overline{x}_i$ and action $\overline{x}_{i+1}$. This gives the cutpoint condition

$$\omega_i = \frac{1}{2}(\overline{x}_i + \overline{x}_{i+1}) - b.$$

Theorem 5.6 (Crawford–Sobel, 1982)**.** *There exists a finite $N^*(b)$ such that for each $N \leq N^*(b)$ there is a N-partition equilibrium: $[0, 1]$ is cut into N intervals $[\omega_0, \omega_1], \ldots, [\omega_{N-1}, \omega_N]$; S reports which interval ω lies in; R plays the midpoint. At boundaries ω_i, the sender is indifferent:*

$$(\omega_i - \bar{x}_i) + b \;=\; \bar{x}_{i+1} - (\omega_i + b) \quad \Leftrightarrow \quad \omega_i = \tfrac{1}{2}(\bar{x}_i + \bar{x}_{i+1}) - b,$$

yielding the difference equation $\omega_{i+1} - \omega_i = \omega_i - \omega_{i-1} + 4b$.

As $b \downarrow 0$, $N^*(b) \uparrow \infty$ and communication becomes arbitrarily fine; as b grows, only babbling survives.

5.5 Dynamic Games

5.5.1 War of Attrition

First let's start with complete information. Two players compete for a prize V. Each pays flow cost 1 per unit time while still "fighting"; the first to quit loses the prize, and the survivor gets V. Players choose stopping strategies. Three classes of SGPE:

(i) Player 1 never stops, player 2 stops immediately.

(ii) The mirror-image.

(iii) **Symmetric MSE:** each player's stopping time T_i is distributed exponentially with rate $\lambda = 1/V$. Check: conditional on having not stopped by t, indifference requires the marginal cost 1 equal the marginal benefit $\lambda \cdot V$ of being the sole survivor in the next instant. So $\lambda = 1/V$.

Proposition 1 (Asymmetric costs)**.** *If player i has flow cost c_i, the symmetric-stationary equilibrium has player i quitting at rate $\lambda_i = c_j/V$. The lower-cost player yields* sooner *in expectation (counterintuitive: her continuation value from fighting longer is higher).*

Now consider that there are assymetric costs. Types: with small prior z_i player i is "behavioral" (never quits); with prob $1 - z_i$ she is rational. Rational players want to mimic behavioral types to build reputation. In continuous-time limit, each rational type concedes at a rate calibrated to keep the opponent indifferent; the war ends at a finite random time T with probability

1 unless one player is behavioral, in which case the other eventually concedes. Posterior reputation updates Bayesian:

$$\frac{d\mu_i(t)}{1-\mu_i(t)} \;=\; \lambda_i\, dt,$$

where λ_i is the rational-type's concession rate. Choosing λ_i to match the rival's indifference gives the "reputational" war-of-attrition prediction: more reputation (smaller z_i of behavioral type) strengthens your bargaining posture.

5.5.2 Global Games

Consider a binary action {Invest (1), Not (0)}. Payoff to investing given fundamental θ and opponent's action:

$$u(1,1;\theta)=\theta, \quad u(1,0;\theta)=\theta-1, \quad u(0,a;\theta)=0.$$

Complete information: if $\theta<0$, "Not" is dominant; if $\theta>1$, "Invest" is dominant; if $0<\theta<1$, both are equilibria (coordination). The symmetric MSNE has probability of Invest $=\theta$.

Suppose each player observes $x_i=\theta+\sigma\varepsilon_i$ with $\varepsilon_i\sim\mathcal{N}(0,1)$ i.i.d., and $\theta\sim\mathcal{N}(0,\tau^2)$ with $\tau^2\to\infty$ (improper prior). Posterior beliefs: $\theta\mid x_i\sim\mathcal{N}(x_i,\sigma^2)$; $x_j\mid x_i\sim\mathcal{N}(x_i,2\sigma^2)$.

Proposition 2 (Global-game selection)**.** *The unique strategy profile that survives iterated elimination of strictly dominated strategies is a cutoff strategy: each player invests iff $x_i\geq\kappa^*$ with $\kappa^*=1/2$ (in the Laplacian limit $\sigma\to 0$).*

Key idea. Suppose agent j uses cutoff k. Agent i's best response cutoff $b(k)$ satisfies $\Pr[x_j<k\mid x_i=b(k)]=\theta$ (so indifference), i.e., $\Phi((k-b(k))/(\sqrt{2}\sigma))=b(k)$. This map is a contraction when iterated; its unique fixed point is $\kappa^*=1/2$, implementing the **risk-dominant** equilibrium of the complete-information game. □

We can also think about the idea of public vs private information. If agents also observe a public signal $y=\theta+\tau\eta$ with common $\eta\sim\mathcal{N}(0,1)$, the cutoff equilibrium is *not* unique when σ/τ is small enough: public information restores multiplicity (Morris–Shin 2004). The sharp condition is

$$\sigma^2/\tau^4 \;<\; 2\pi,$$

guarantees uniqueness; otherwise multiple cutoff equilibria emerge.

5.6 Exercises